



Dept. of Environmental Science and Engineering
Jatiya Kabi Kazi Nazrul Islam University
Trishal, Mymensingh 2220, Bangladesh

Syllabus for B.Sc. (Engg.) Degree in Environmental Science and Engineering

Session 2021-2022

Summary of Credit Hours/Semester

Year	Semester	Classes (Hours/Week)	Credits
1	1	26.0	21.5
	2	24.0	21.5
2	1	28.0	22.0
	2	20.0	19.0
3	1	24.0	20.0
	2	19.0	19.5
4	1	22.0	19.0
	2	17.0	19.5
Grand Total	8	180	162.0

Distribution of Credits among Major Fields

Major Fields	Total Credits	Relative Percentage (%)
Physics	3.5	2.16
Chemistry	15.0	9.26
Mathematics	6.0	3.70
Humanities	12.0	7.41
Environmental Science and Engineering	101.5	62.65
Related Engineering**	24.0	14.81
Grand Total	162.0	100.00

Related engineering courses include courses on Computer Science and IT, Engineering Drawing, Survey and Settlement, Soil Mechanics, Environmental Engineering, Integrated Water Resource Development and Engineering, Environmental Risk Analysis and Modeling, Public Health Engineering, and Environmental Biotechnology.

Year-wise Distribution of Courses

1st Year 1st Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 1101	Introduction to Environmental Science and Engineering	3	0	3
ESE 1103	Fundamental of Physics	2	0	2
ESE 1102	Fundamental of Physics –Lab	0	3	1.5
ESE 1105	Fundamental of Chemistry	2	0	2
ESE 1104	Fundamental of Chemistry-Lab	0	3	1.5
ESE 1107	Fundamental of Earth Sciences	2	0	2
ESE 1106	Fundamental of Earth Sciences –Lab	0	3	1.5
ESE 1109	Mathematics-I	3	0	3
ESE 1111	Nazrul Studies	3	0	3
ESE 1113	Bangladesh Studies	2	0	2
Total Credits				21.5

1st Year 2nd Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 1201	Geology and Geomorphology of Bengal Basin	2	0	2
ESE 1203	Environmental Earth Sciences	2	0	2
ESE 1202	Environmental Earth Sciences - Lab	0	3	1.5
ESE 1205	Environmental Biology	2	0	2
ESE 1204	Environmental Biology - Lab	0	3	1.5
ESE 1207	Mathematics-II	3	0	3
ESE 1209	Computer Science and IT	2	0	2
ESE 1206	Computer Science and IT - Lab	0	3	1.5
ESE 1211	Sociology for Engineers	2	0	2
ESE 1213	Communicative English	2	0	2
ESE 1208	Field work	0	0	1
ESE 1210	Viva Voce	0	0	1
Total Credits				21.5

2nd Year 1st Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 2101	Hydrology and Hydrogeology	3	0	3
ESE 2103	Oceanography and Limnology	2	0	2
ESE 2105	Ecology	2	0	2
ESE 2102	Ecology- Lab	0	3	1.5
ESE 2107	Environmental Microbiology	2	0	2
ESE 2104	Environmental Microbiology- Lab	0	3	1.5
ESE 2109	Survey and Settlement	2	0	2
ESE 2106	Survey and Settlement-Lab	0	3	1.5
ESE 2111	Soil Mechanics	2	0	2
ESE 2108	Engineering Drawing Lab.	0	3	1.5
ESE 2113	Statistics for Environment	3	0	3
Total Credits				22.0

2nd Year 2nd Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 2201	Forest and Wildlife Management	3	0	3
ESE 2203	Environmental Geochemistry	2	0	2
ESE 2202	Environmental Geochemistry- Lab	0	3	1.5
ESE 2205	Meteorology and Climatic Process	2	0	2
ESE 2204	Meteorology and Climatic Process –Lab.	0	3	1.5
ESE 2207	Coastal Environment and Blue Economy	3	0	3
ESE 2209	Environmental Biochemistry	2	0	2
ESE 2211	Biodiversity Conservation	2	0	2
ESE 2206	Field Work	0	0	1
ESE 2208	Viva Voce	0	0	1
Total Credits				19.0

3rd Year 1st Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 3101	Environmental Pollution	3	0	3
ESE 3102	Environmental Pollution-Lab	0	3	1.5
ESE 3103	Soil Science	3	0	3
ESE 3105	Environmental Health and Occupational Safety	2	0	2
ESE 3107	Energy and Environment	2	0	2
ESE 3109	Remote Sensing and GIS	3	0	3
ESE 3104	Remote Sensing and GIS –Lab.	0	3	1.5
ESE 3111	Solid Waste Management	3	0	3
ESE 3113	Environmental Ethics, Law and Policy	2	0	2
Total Credits				20.0

3rd Year 2nd Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 3201	Environmental Hazard and Disaster Management	3	0	3
ESE 3203	Waste and Waste Water Treatment	2	0	2
ESE 3202	Waste and Waste Water Treatment -Lab.	0	3	1.5
ESE 3205	Environmental Sustainability	2	0	2
ESE 3207	Environmental Economics	2	0	2
ESE 3209	Integrated Water Resources Development and Engineering	3	0	3
ESE 3211	EIA and Environmental Auditing	2	0	2
ESE 3213	Agriculture and Environment	2	0	2
ESE 3204	Field work	0	0	1
ESE 3206	Viva voce			1
Total Credits				19.5

4th Year 1st Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 4101	Flood Control and Watershed Management	2	0	2
ESE 4103	Environmental Chemistry	2	0	2
ESE 4102	Environmental Chemistry- Lab.	0	3	1.5
ESE 4105	Environmental Engineering	3	0	3
ESE 4107	Global Warming, Climate Change and Adaption	3	0	3
ESE 4109	Natural Resources Management	2	0	2
ESE 4111	Environmental Risk Analysis and Modeling	2	0	2
ESE 4104	Environmental Risk Analysis and Modeling-Lab	0	3	1.5
ESE 4113	Research Methodology	2	0	2
Total Credits				19.0

4th Year 2nd Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 4201	Water Supply and Sanitation	3	0	3
ESE 4203	Environmental Planning and Design	3	0	3
ESE 4205	Environmental Biotechnology	2	0	2
ESE 4207	Industrial Ecology	2	0	2
ESE 4209	Public Health Engineering	2	0	2

ESE 4211	Water Pollution and Quality Management	2	0	2
ESE 4202	Water Pollution and Quality Management -Lab	0	3	1.5
ESE 4204	Research/ Project Works	0	0	3+1 (Report + Presentation)
Total Credits				19.5
Grand Total Credits				162.0

Detail Course Contents for Four Years B.Sc. (Engg.) Degree in Environmental Science and Engineering

Academic Year: 2021-2022

1st Year 1st Semester

ESE1101. Introduction to Environmental Science and Engineering Credits: 3; Contact Hours: 3 Hours/Week

Introduction: Definition, scope and guiding concept on environmental science, Layers of soil and atmosphere (hydrosphere, biosphere, Lithosphere etc.) Climate factors (light, temperature, precipitation, humidity, atmosphere and air)

Major environmental problems: Green House effect, Ozone layer depletion, Global warming and climate change, agricultural pollution and deforestation, Sea level rise, El-nino, and La-nina, Alzheimer, Acid rain, natural disaster in Bangladesh(flood, earthquake etc)

Environmental conventions and protocols: background and definition; important international conventions and associated protocols related to environmental protection

Environmental Resource: Resources: renewable, potentially-renewable and non-renewable resources; Types of Resources: Land, Water, Forest, Minerals, Agriculture, Wild life and Energy; Status of resources and their proper management.

Tools and standards of environmental management: water quality standards in Bangladesh; environmental impact assessment; precautionary principle; polluter pays principle.

Principles and application of Environmental Technology: mass balance, conservation, energy; application of environmental technology for pollution management.

Recommended References

- Anderson, J. M. Ecology for Environmental Sciences: Biosphere, Ecosystems and Man. Resource and Environmental Sciences Series.
- Stanley E. Manahan: Environmental Science and Technology; Third Edition, CRC Press, Taylor and Francis Group.
- Stanley E. Manahan; Environmental Science and Technology: a sustainable approach to green science and technology; 2nd Edition By CRC Press, Taylor and Francis Group.
- Stanley E. Manahan; Environmental Science and Technology: LEWIS PUBLISHERS, New York.
- Dawei Han: Concise Environmental Engineering
- Bernard, J., N. Richard and T. Wright. Environmental Science (6th ed.). Prentice Hall.
- Mukherjee, S. and H. Ghosh. Environmental Studies, Published by Books and Allied (P) Ltd. Kolkata, India.
- Raven and Berg. Environment (3rd Ed.). Harcourt College Publishers.
- Enger ED and Smith BF (1995) *Environmental Science – a study of interrelationship*, WmC Brown Publisher
- Jackson ARW and Jacson JM (1996) *Environmental Science*, Longman

ESE1103. Fundamental of Physics Credits: 2; Contact Hours: 2 Hours/Week

General Properties of Matter: *Structure of matter*- classification of solids; amorphous, crystalline, binding energy and atomic separation in equilibrium in solid, different types of bonds in crystals, lattice, basis, crystal, unit cell, packing fraction of sc, bcc crystals, X-ray diffraction, Bragg's law. *Elasticity*- Hook's law, Shear and torsion of a cylinder, Elastic constants and use of their interrelations. Bending of beam (cantilever). *Surface Tension*- Molecular theory of surface tension, total surface energy, excess pressure inside a curved surface, the capillary ascent and its use, Molecular force, total surface energy, pressure on a curved surface. Determination of surface tension of water and mercury. *Viscosity and fluid dynamics*- Newton's law of viscous flow, Poiseuille's equation, the equation of continuity for streamline flow, Bernoulli's equation and its application. *Viscosity*: Newton's law of viscous flow. Flow of liquid through a tube. Poiseuille's equation, energy possessed by a liquid; Bernoulli's theorem.

Heat and Thermodynamics: Ideal gas equation, fundamental postulates of kinetic theory of gases, kinetic interpretation of temperature. First law of thermodynamics and its applications, different kind of thermodynamical

changes, interrelation between pressure, volume and temperature during those changes, specific heats of ideal gases, equipartition energy. Second law of thermodynamics reversible and irreversible processes, heat engine and Carnot's cycle, Carnot's theorem, thermodynamic scale of temperature, entropy and second law, entropy and disorder. Radiation of heat, black body and body and radiation, laws of radiation, planck's constant.

Electricity and Magnetism: Current and resistance, Kirchhoff's laws on distribution of current, its applications R-C circuit. Magnetic induction due to current, Ampere's law, Biot-Savart law, magnetic induction due to straight, circular and solenoidal current, magnetic properties of matter. Electromagnetic induction. Henry-Faraday's law, inductance, L-R circuits. Alternating current and L-R circuits, impedance and reactance.

Nuclear physics and electronics: *Nuclear physics*- Atomic and nuclear structure, radioactivity, decay law, half life. nuclear fission and fusion, use of radio isotopes. *Electronics*- Semiconductors, transistors and integrated circuit and their uses.

Recommended References:

- D. S. Mathur. Properties of Matter
- Resnick, R and Halliday, D. Physics for Science and Engineering students part I & II.
- Sears, F.W. and others.: University Physics part I and II
- Newman F.H. & Searey VHL. The general properties of matter.
- Edward-Arnold Kenneth, Wark : Thermodynamics.

ESE1102. Fundamental of Physics- Lab

Credits: 1.5; Contact Hours: 3 Hours/Week

Determine Y (Young's modulus) by Searle's method.

Determination of the viscosity of a given liquid and its variations with temperature.

Determine surface tension by capillary rise.

Determine modulus of rigidity by dynamic method.

Determination of the thermal conductivity of a good conductor by Searle's method.

Determination of the thermal conductivity of a bad conductor by LEE'S method.

Determine 'g' by Kater's pendulum. To verify Biot-Savart law and Tangent law.

Determine H by Vibration magnetometer. Compare magnetic moments of two bar magnets in TanA and TanB by position by deflection method.

Recommended References:

- Practical Physics. Dr. Gius Uddin Ahmed and Md. Shahabuddin, Hafiz Book Centre
- Practical physics , Millikan, Robert Andrews, 1868-1953; Gale, Henry Gordon, 1874-1942; Pyle, Willard R

ESE1105. Fundamental of Chemistry

Credits: 2; Contact Hours: 2 Hours/Week

The States of Matter: The Gaseous States: gas laws, kinetic theory of gases, distribution of molecular velocities, molecular diffusion and effusion, deviations from ideal gas behavior, the Van der Waals equation of state; The Liquid State: liquefaction of gases, the critical state, determination of critical constants, vapor pressure of liquid, surface tension; The Solid State: properties of solids, crystalline and amorphous solids, the crystal system.

Atomic Structure: The atom, Nuclear charge and atomic number; Rutherford and Bohr's atomic model Quantum Numbers, Pauli's exclusion principle; Ionization energies, electron affinity, Wave nature of electrons.

Periodic Table of Elements: Periodic classification of elements; Periodic properties

Chemical bonds: Electronic concept of chemical bonds, Types of bonds; Modern concept of covalent bond.

Oxidation and Reduction: Electronic concept, Oxidation state and oxidation number, writing of equations involving oxidation-reduction reactions

Acid/Base Equilibrium: Modern concept of acids and bases; Ionic product of water

Chemistry of Solutions: Definition, Ways of expressing concentration; Theory of dilute solutions, Ideal solution, Raoult's law, Colligative properties

Colloids: Properties of colloidal system; Concept of pH and buffer

Thermodynamics: Introduction, Laws of thermodynamics; Reversible and irreversible processes, Thermochemistry

Recommended References

- Principles of Physical Chemistry, M.M. Haque and M.A. Nawab (Student Publications).
- Elements of Physical Chemistry, S. Glasstone and D. Kews (McMillan & Co. Ltd).
- Physical Chemistry, P.W. Atkins.
- General Chemistry, D.D. Ebbing..
- Introduction to Modern Inorganic Chemistry, S.Z. Haider.
- Basic Inorganic Chemistry, Cotton and Wilkinson, 4th Edn.
- Organic Chemistry, R.H. Morrison and R.N. Boyd, Prentice-Hall, Inc.

Maintenance of chemistry lab- Practice of volumetric analysis.

Preparation of Solutions- Preparation of normal and molar solution of common acid and bases, preparation of normal solution of oxidizing and reducing reagents (e.g., potassium dichromate, sodium thiosulfate, ferrous ammonium sulfate), preparation of dilute solutions, series dilution and strengths calculation of primary standard substances.

Acid-Base Reaction- Standardisation of base against standardise acid, standardisation of acid against standardise base.

Oxidation-Reduction Reaction- Determination of ferrous iron, ferric iron and total iron by oxidation with standard potassium dichromate solution.

Recommended reference

Eaton AD, Greenberg AE, and Cleceri LS (1996) Standard Methods. APHA.

Introduction: Introduction, Importance of earth sciences in environmental science, elemental, mineralogical and rock composition of the earth crust.

The Universe, Solar System and the Planet Earth: Big-Bang theory, Origin of the Earth and the solar system, interior of the Earth.

Earth's Surface Processes: Weathering, erosion and denudations; the features of Continents and Oceans; Isostasy, Earthquakes and Volcanism.

Plate Tectonics: Basic concepts of plate tectonics- evidences for continental drift, type of plate boundaries,.

Mineralogy: Basic knowledge of mineralogy, crystallography; crystal system, physical properties and classification of minerals, common rocks and soil forming minerals.

Petrology: Basic knowledge of rock, rock cycle, major rock types and their formation, classification and description, kinds of soil development from different rocks.

Recommended References:

- Skinner, B.J. & Porter, S.C., 1995, The Dynamic Earth, 3rd Edn, John Wiley & Sons, Inc., New York.
- Holmes, D.L., 1978, Holmes Principles of Physical Geology, ELBS and Thomas Nelson & Sons Ltd., Nairobi.
- Judson, S. and Kauffman, M.E, 1990, Physical Geology, Prentice-Hall Inc., New Jersey.
- Bradshaw, M.J. et al. The Earth's Changing Surface
- Leet, L.D. et. al. Physical Geology
- Montgomery, C.W. Physical Geology

Identification of common rock forming Minerals on hand specimen - Study of physical properties of minerals.

Laboratory works based Petrology and Mineralogy- Study and Identification of common Igneous, Sedimentary and Metamorphic rocks in hand specimen.

Differential Calculus: Function, limit of a function, continuity and differentiability of functions, successive differentiation, physical applications of first and second order derivatives, Leibnitz's theorem, maxima and minima for functions of one and two variables, functions of several variables, partial differentiation, tangent, normal and curvature.

Integral Calculus: Review of indefinite integrals, definite integrals, properties of definite integrals, gamma and beta functions, integration by reduction, rectification, quadrature, surface areas and volumes of revolution.

Matrices: Definitions of matrices, solution of linear equations by matrices, rank of a matrix.

Recommended References

- Matin, M. A. and B. B. Chakravarty. Differential Calculus
- Matin, M. A. and B. B. Chakravarty. Integral Calculus
- Ross, S. L. Differential Equation
- Das & Mukherjee: Differential Calculus
- Das & Mhukerjee: Integral Calculus
- Shahidullah & Bhattacharjee: A Text Book on Higher Algebra & Trigonometry

নজরুলের জীবন ও দর্শন

কাব্য : অগ্নি-বীণা

উপন্যাস : মৃত্যুক্ষুধা

ছোটগল্প : ব্যথার দান গল্পগ্রন্থ থেকে: ব্যথার দান, হেনা, বাদল-বরিষণে; রক্তের বেদন গল্পগ্রন্থ থেকে: রক্তের বেদন, বাউগেলের আত্মকাহিনী, রাফুসী ; শিউলিমালা গল্পগ্রন্থ থেকে: পদ্ম-গোখরো, অগ্নি-গিরি, শিউলিমালা

গীতি : অঞ্জলি লহ মোর সঙ্গীতে; এ কি অপরূপ রূপে মা তোমায় হেরিনু পল্লী-জননী; ও ভাই খাঁটি সোনার চেয়ে খাঁটি; খেলিছ এ বিশ্ব লয়ে; চল চল; তুমি কি আসিবে না; পদ্মার ঢেউ ও; লাইলী তোমার এসেছে ফিরিয়া; শাওন রাতে যদি স্মরণে আসে মোও; শুকনো পাতার নূপুর পায়ে .

প্রবন্ধ : ডায়ারের স্মৃতিস্তম্ভ; বাঙলা সাহিত্যে মুসলমান; কালা আদমীকে গুলি করে মারা; সত্য-শিক্ষা; জাতীয় শিক্ষা; জাতীয় বিশ্ববিদ্যালয়; রাজবন্দীর জবানবন্দী; মন্দির ও মসজিদ ; হিন্দু-মুসলমান; বর্তমান বিশ্ব-সাহিত্য.

অভিভাষণ : যৌবনের গান; যদি আর বাঁশী না বাজে.

নাটক : পুতুলের বিয়ে;ঝিলিমিলি.

পত্র : প্রিন্সিপাল ইব্রাহীম খাঁ লিখিত 'চিঠির উত্তরে'; বেগম শামসুন নাহার মাহমুদকে লিখিত পত্র .

সহায়ক গ্রন্থপঞ্জী :

- অরুণকুমার বসু, নজরুল জীবনী
- আজহারউদ্দীন খান, বাংলা সাহিত্যে নজরুল
- আবদুল মান্নান সৈয়দ, নজরুল ইসলাম : কবি ও কবিতা
- করুণাময় গোস্বামী, নজরুল-গীতি প্রসঙ্গ
- ধ্রুবকুমার মুখোপাধ্যায়, নজরুল ইসলাম : কবি মানস ও প্রতিভা
- নারায়ণ চৌধুরী, নজরুল চর্চা
- বিশ্বজিৎ ঘোষ, নজরুল মানস ও অন্যান্য প্রসঙ্গ
- মোহাম্মদ আবদুল কাইউম, নানা প্রসঙ্গে নজরুল
- মোবাক্কের আলী, নজরুল প্রতিভা
- মাহবুবুল হক, নজরুল তারিখ অভিধান
- রফিকুল ইসলাম, কাজী নজরুল ইসলাম : জীবন ও সৃজন
- রাজিয়া সুলতানা, কথাশিল্পী নজরুল
- সুশীলকুমার গুপ্ত, নজরুল-চরিতমানস
- সৌমিত্র শেখর, নজরুল : আন্তর্জাতিক সম্প্রীতি এবং শিল্পীর বোধ

Course Outlines:**History of Bangladesh:**

People of Bangladesh, Origin and Anthropological Identity, Territories of Bangladesh in Ancient Time, Bangladesh During the Muslim Rule, British Colonial Rule in the Indian Sub-Continent, the Pakistan Movement and the Birth of Pakistan, Language Movement, 1948-52, Juktafront Elections, 1954 and Six Point Movement-1966, Mass Upsurge of 1969 and Elections, 1970, Liberation War, 1971 and Birth of Bangladesh.

Physical and Natural Resources of Bangladesh:

Physical Features of Bangladesh, Location, Climate, and Major Physiographic Units, Surface Water Inflow and River Systems of Bangladesh, Pattern of Agriculture and Type of Forest in Bangladesh, Mineral Resources and Industrial Set-up in Bangladesh, Density and Distribution of Population in Bangladesh.

Political Administrative and Legal Environment of Bangladesh:

Forms of Government and State Mechanisms (Executive, Legislature, Judiciary) of Bangladesh, Constitution of Bangladesh and its Amendments, Administrative Structure of Bangladesh, Political Parties of Bangladesh, Different Political Regimes of Bangladesh, Foreign Policy of Bangladesh.

Socio-Economic Environment of Bangladesh:

Social Structure and Social Stratification in Bangladesh, Population Migration from Rural to Urban Areas in

Course Objectives:

The major objective of the course is to introduce the students to the history of Bangladesh, to understand present Bangladesh in the light of history and to provide them with basic knowledge of current politics and economy of the country. This course will expand students understanding of complex interconnection of historical events which led to the formation of Bangladesh, current trend in political and economic development thereby improving their critical thinking. Having completed the course, the students will be able to:

- Identify specific stages of Bangladesh's political history from the ancient periods to the emergence of Bangladesh.
- Analyze the rich history, culture and heritage of Bangladesh.
- Critically analyze how different constitutional bodies and socio-political institutions operate and how their behavior impact on political governance.
- Explain the economy and pattern of economic and political changes through qualitative and quantitative analysis.

Assessment:

Midterm/Assignments/Presentation	: 30%
Attendance	: 10 %
Semester Final	: 60%

Text Books and Reference Book:

- Bangabandhu Sheikh Mujibur Rahman: The Unfinished Memoirs.
- Sirajul Islam : Bangladesh: National Encyclopedia of Bangladesh (Latest Edition)
- Sirajul Islam: History of Bangladesh, Social and Cultural History, Political History, Economic History.
- Ahmed, A F Salauddin: Bangladesh National Culture and Heritage, An Introductory Reader Edited.
- M.A Ahad: Bangladesh Economy.
- Mahab Khan: Bangladesh society, Politics and Bureaucracy.

1st Year 2nd Semester

ESE1201. Geology and Geomorphology of the Bengal Basin

Credits: 2; Contact Hours: 2 Hours/Week

Introduction: Geographic Location and Geological Position of the Bengal Basin, population, physiography, hydrology, climate and vegetation, mountains, glaciers, waterfall, rivers, lakes, earthquake, volcanoes.

Quaternary geomorphology: Pleistocene uplands, Holocene sediments.

Geologic evolution of the Ganges-Brahmaputra-Meghna Delta: Early and late stages of development; Shifting of the Ganges Delta with time; Origin of the Himalayan and Gangetic flood plains.

Tectonic evolution and basin-fill history of the Bengal Basin: Physiography and hydrography of the Bengal delta - major subdivisions; Tectonic framework of Bangladesh, Stable Platform; Hinge Zone; Bengal Foredeep - Calcutta (Kolkata)-Mymensingh Gravity High; Barisal-Chandpur Gravity High; Sylhet (Surma) Trough; Hatiya Trough; Tertiary Foldbelts of the Bengal Basin; Offshore troughs and sub-basins.

Geotectonic subdivisions of the Bengal Basin: Stratigraphic successions of the geotectonic provinces of the Bengal Basin

Recommended References:

- Curray, J.R. & Moore, D.G., 1974, Sedimentary and Tectonic Processes in the Bengal Deep-Sea Fan and Geosyncline; In: Burke, C.A. & Drake, C.L. (eds.)
- The Geology of the Continental Margins, Springer-Verlag, Heidelberg, New York, p. 617-627.
- Evans, P., 1934, The Tectonic Framework of Assam; Geological Society of India Journal, vol. 5, p. 80-86.
- Khan, F.H., 1991, Geology of Bangladesh; The University Press Ltd., Dhaka.
- Reimann, K.U., 1993, Geology of Bangladesh; Gebrüder Borntraeger, Berlin
- James S. Monroe & Reed Wicander. 2006, *The changing earth exploring geology and evolution* 4th edition, Books/Cole Cengage Learning
- Mark J. Crawford. 1998, *Physical Geology*, 1st edition, Cliffs notes Ins. USA
- Plummer C.C., Carlson D.H. and Mc Geary. 200, *Physical Geology*, 11th Edition, McGraw Hill

ESE1203. Environmental Earth Sciences

Credits: 2; Contact Hours: 2 Hours/Week

Introduction to Earth Sciences: Definitions of Earth Sciences and Environmental Science, relationship between Earth Sciences and Environmental Science.

Weathering and Soil: Weathering, erosion, transportation and deposition of sediments and sedimentary rocks; types of weathering, factors influencing weathering, effects of weathering; factors affecting weathering; soil and soil forming factors,

Mass wasting: Definition; Selected mass wasting, their cause, and deaths; factors influence mass wasting; Classification of mass movement and their characteristics.

Stream action: Water on Earth; running water; Drainage Basins; Drainage Patterns; factors affecting stream; stream erosion; floods; Lakes; the life cycle of lakes.

Wind action and deserts: Wind transportation, wind erosion landforms; types of wind deposits; Deserts landforms; dunes.

Crustal Deformation of Rock: Geologic structures (folds, faults and joints) - definition and classification, Mountain Building and their environmental significance.

Geological Time: Fundamental Concepts of Geology and principles, unconformities, and Development of Geological Time Scale.

Fossils, Evolution and Extinction: Definition of fossil, types of preservation, conditions that favor preservation of fossils, types of evolution, reasons of extinction.

Recommended References:

- Banfar, M.M, ATextbook of Geology, 1981, Standard publisher distributors, Delhi.
- S. M . Sirajul Haque, Geology and Soil Science IFESCU and USDA
- Hussain, MS Soil Classification (with special reference to the soils of Bangladesh) 1992, University of Dhaka.
- Brammer, H. Soil Survey Project Bangladesh. 1971. Soil Resources. AGL;SF/PAK
- Brady, N.C The nature and properties of soil, 1996. Prentice hall, India, New Delhi.
- Spencer EW (1983) Physical Geology. Addison-Wesley Pub Co
- Emiliani C (1992) Planet Earth. Cambridge Univ. Press
- Krishnan MS (1982) Geology of India and Burma. CBS Pub and Distributors, India
- Reimann KU (1993) Geology of Bangladesh. Gebrüder Borntraeger, Berlin
- Khan FH (1991) Geology of Bangladesh. Wiley Eastern, New Delhi
- Rashid H er (1991) Geography of Bangladesh. Univ Press Ltd, Dhaka
- Rogers JJW and Feiss PG (1998) People and the Earth – basin issues in the sustainability of resources and environment. CUP

ESE1202. Environmental Earth Sciences-Lab**Credits: 1.5; Contact Hours: 3 Hours/Week**

Environmental surveying, topo-sheet and map reading, map projection, subsurface geologic data interpretation.

Hand-specimen and Microscopic Study of Clastic and Nonclastic rocks. Thin-section and Grain-size Analyses; Dry- and Wet-Sieve Methods; Representation and Statistical Analyses of Grain-size Parameters and their Interpretation; Heavy-mineral Separation; Microscopic Study of Light and Heavy minerals, and facies analysis

Studying physical, chemical and biological properties of soils and soil profile; determinations of moisture content, salinity and pH of soil.

Microscopic technique for identification of microfossils (dinocysts, pollen, chlorococcales) and foraminiferal shell and application of micropaleontology on climate/paleoenvironment evaluation.

Determining the environmental behavior of water-rock interaction-analysis and interpretation of Hydrology and Hydro-geological data.

Recommended References:

- Folk, R.L. Petrology of Sedimentary Rocks.
- Milner, H.B. Sedimentary Petrography (vols. I & II.)

ESE1205. Environmental Biology**Credits: 2; Contact Hours: 2 Hours/Week**

Biological Principles of life- Definitions of life, General features of living system (Chemical uniqueness, complexity)

Unit of life- Definition and classification of cell, Prokaryotic and eukaryotic plant cells, comparison between the ultra-structure of eukaryotic plant and animal cells, Mitochondria and Respiration, Chloroplasts and photosynthesis, physiology of cells (energy, enzymes and metabolism), Definition and structure of DNA & RNA, chromosome and cell division.

Microscopic life- introductory discussion on diversity, morphology and physiology of microscopic life (Monera, protista, Fungi)

Kingdom- Animal kingdom: Classification, diversity, morphology, physiology and multiplication of environmentally significant animal phyla such as Protozoa, porifera, Coelenterata, Annelida etc).

Plant Kingdom: Introduction, Classification and diversity of plant kingdom, physiology and multiplication of plants, Environmental factors influencing plant growth, role of plant kingdom in environmental Equilibrium.

Biodiversity in Bangladesh- Faunal and floral Diversity- relation of biology and environment.

Recommended References

- Atlas, R.M. & Bartha, R.2000. Microbial Ecology-Fundamentals and applications. 4th ed., the Benjamins, Cummings Publ. Co., U.S.A., Canada.
- Chapman, J.L. & Reiss, M.L.1999. Ecology; Principles and applications, 2nd ed. Cambridge Univ. Press, Cambridge, U.K.
- David McFarland, 1985. Animal Behaviour, Sychobiology, Ethology and Evolution, ELBS, Oxford.
- A textbook of Biotechnology. R.C. Dubey. 2nd edition; 1995. S. Chand and Company Ltd. New Delhi.

ESE1204. Environmental Biology- Lab**Credits: 1.5; Contact Hours: 3 Hours/Week**

Field survey on Biology for Environmental Sciences (protected areas, e.g., Eco-park); Introduction to laboratory; Microscopic Technique; Preservatives, Stains and Mounting Media; Study of algae, fungi and lichen; Study of bryophytes and pteridophytes; Demarcation of phytoplankton and zooplankton; Study of bacterium: Staining technique; Preparation of herbarium sheet and preservation of fauna; Identification and characterization of Chondichthis and Osteichthis; Preparation of field and laboratory note book.

ESE 1207. Mathematics-II**Credits: 3; Contact Hours: 3 Hours/Week**

Differential Equations: Definition, types of differential equations, formation of differential equations, solution of first order and first degree equations, solution of linear differential equations of second and higher orders with constant coefficients, method of variation of parameters, solution of homogeneous linear equations.

Vector calculus: Scalar & vector functions, gradient of a scalar function, curl & divergence of a vector function, Green's theorem, Gauss's divergence theorem and Stokes theorem.

Numerical analysis: Introduction, solution of algebraic and transcendental equations, interpolation and extrapolation, numerical differentiation and integration, numerical solution of ordinary differential equations.

Recommended References:

- J Edwards; Differential Calculus
- Das & Mukharjee: Differential Calculus
- Das & Mukharjee: Integral Calculus
- M R Spiegel: Advanced Calculus
- S. L. Ross: Differential Equations
- Frank Ayres: Differential Equations

ESE1209. Computer science and IT**Credits: 2; Contact Hours: 2 Hours/Week**

Introduction to Computer: a) Definition, Application, Generations of computers, Basic concepts of computing system, Anatomy of digital computer, Classification of computer, b) Bit-Byte concepts, Computer memory, Keyboard instruction, c) Introduction to windows operating system, Single/multi-user, Multitasking, Windows explorer, d) Create folder, Copy folder, Rename folder/file, Find file or folder.

MS Word: a) Definition of word processor, File management, Save, Save As, Save to floppy. Disk/other storage device, b) Edit: Cut, Copy, Paste, Undo, Redo, c) Formatting, Alignment, Character Spacing, font, Superscript, Subscript, Line, Spacing, Insert Symbol, d) Insert Page Break, Insert Page number, Insert Date and Time, Paragraphs, Indents, e) Bullet and Numbering, Header and footer, Border and Shading, Change Case, f) Tab Settings, Spelling and Grammar, Page Setup, Print Preview, Print, g) Auto Correct, Format Painter, Hyperlink, Paste Special (Paste Link), Go to Specific Page, Find and Replace, h) Working with Picture, Drawing and Art, Using Typing Tutor, Bangla font, i) Tables, Drop Cap, j) Create News Paper Columns, Insert Column Break, Envelops, Labels.

MS Power Point: a) Introduction, Slide Definition; b) Create, Delete Slides, Navigate Views (Slide, Outline, Sartre), Change the order of Slides, Apply Add Auto design template, Apply Back ground, Numbering bullets Add graphical Bullet, Add Slide transitions, start Slide Show; c) Animate text and Objects, Preview Presentation, in black and white, Print Slider in a Variety of Formats, Using pen in Presentation

MS Excel: a) Introduction to Excel, delete, Copy, Cut Text, Delete Cell, Insert and Delete Row and Column, b) Changing Row height and Column Width, Select Different Cells, Rows, Columns, c) Freeze and Un Freeze Pane, Paste Special, Auto fill, Calculations, d) Functions (IF, AND, PMT, FV etc.), e) Conditional formatting, Consolidate data, f) Header and Footer, Cell formatting, page Setup, Sorting, Data Filtering, g) Charts

Internet and Web Site Related Topic: Internet service, E-mail, E-Commerce and Basic Networking Concept

Recommended References

- Norton, P. Introduction to Computers.
- Rajarcman, V. Fundamentals of Computers.
- Subramanian, N. Introduction to Computers.
- Sinha, P. K. Computer Fundamentals.

ESE1206. Computer Science and IT – Lab**Credits: 1.5; Contact Hours: 2 Hours/Week**

To be designed on the basis of theoretical classes.

Operating System: Windows 10 or later version; **Application Program:** Microsoft office; **Photo editing; Visual basic**

ESE1211. Sociology for Engineers**Credits: 2; Contact Hours: 2 Hours/Week**

Nature and scope of sociology: Development of sociology as a science; Relation of sociology with other disciplines; Importance of sociology from environmental science point of view, Society; Community; Group; Association; Institution; Culture and Civilization.

Social Structure: Concept, views and elements of social structure; Social stratification and its different forms; Social mobility; Social Institutions; its Characteristics and role of Social institutions and Social Change: Concept, sources and consequences of social change; Major theories of social change; Urban ecology.

Culture and Civilization: Social Problems Meaning and nature of social problem; Major social problems in the context of Bangladesh - Population explosion, Poverty; Slum, Beggary, Woman torture and Dowry.

Environment and Social Ecology: Population and ecology; Contemporary major environmental issues- Industrialization, Urbanization, Land Use and Agriculture, Energy Consumption

Social Control: Definition and types of social control, Agencies of Social control

Environmental Awareness: Environmental awareness and its implications need for Environmental awareness, Awareness creation as a tool for disaster management, Environmental awareness creation and disaster management

Informal Environmental Education: Awareness Development: Individual and Public, Social Mobilization and Leadership, Children and Gender in Informal Environmental Education, Coordination and Information Management

Environmental Education Movement: Organizational Development in National and International level, Youth in Action, Environmental Education and Economics, Environmental Education and Politics, Role of Govt. and NGOs

Recommended References

- Bottomore, T. B., 1975, Sociology: A Guide to Problems and Literature, Blackie and Son (India) Ltd.
- Hanningan, J. A., 1995, Environmental Sociology, Routledge, London
- Rahman, Atiqur et al, Environment and Development in Bangladesh
- Robertson, I., 1977, *Sociology*, Worth Publishers, Inc. New York
- WL Filho, J Padgham, JU Ahmed, A.H. Khan and A.H. Ullah, 1994, Communicating Environmental risks: Education as tool for Environmental Disaster Management, CES, Jahangirnagar University.

ESE1213. Communicative English**Credits: 2; Contact Hours: 2 Hours/Week**

Basic Grammar: Sentence sense, Tense, Subject verb agreement, Rules of Using the correct form of the verb, Preposition, Phrasal verbs, Idioms of phrase etc.

Reading Comprehension: Skimming, Scanning, inference, predicting

Development of Writing: Skill-Planning, outlining (Drafting), Revising, Editing, Proof reading, Vocabulary Building, Precise/ Summarizing, Essay writing (with and without clues), Report writing, Preparing resume, Letter/ application writing (Formal/Informal).

Paragraph Development: Paragraph structure, Transitional devices connectives

Case study: Oral presentation, Practice dialogue (Written and oral e.g. on an interview situation)

Recommended References

- Bazerman. Writing Skills Handbook.
- Bentley, T. J. Report Writing in Business.
- Chowdhury, M. Y. A. and M. M. Hossain. Advanced Learners Functional English.
- Seely, J. The Oxford Guide to Writing and Speaking.
- Tailor, C. Advancing Language Skills.
- Ullah, M. K. A Potential English Grammar and Composition.
- Taylor C (1978) Advance Language Skills. The University Press Ltd, Dhaka

ESE1208. Field work**Credit: 1.0**

To be introduced different types of environment, resources and natural diversity in Bangladesh through fieldwork at different places and/or related organizations. Each student shall require submitting a report and presenting a seminar group-wise according to fieldwork.

ESE 1210. Viva voce**Credit: 1.0**

Comprehensive oral examination of the courses taught.

ESE 2101. Hydrology and Hydrogeology

Credits: 3; Contact Hours: 3 Hours/Week

Introduction: Definition, Hydrologic Cycle, Importance of Hydrology, Water Balance, Energy Budget, metrological parameters affecting hydrologic cycle, water shed, hydrologic equation.

Precipitation and Water losses: Types and forms, causes and measurements of precipitation and its analysis;

Infiltration, methods determining Infiltration,

Evaporation and transpiration: Evaporation, evaporation as a process, Factors affecting evaporation, methods determining evaporation and evapotranspiration, transpiration, measurement of evaporation, estimation of evaporation.

Run off and flooding: Sources, Catchment characteristics, Factor affecting runoff and runoff estimation Hydrologic Instrumentation, Drainage Patterns; Stream flow Velocity; River-Stage and Discharge; Floods.

Hydrograph: Definition, Hydrograph separation, Unit Hydrograph, Coastal-plain Hydrology.

Storage and groundwater: Origin and occurrence of groundwater, rock properties affecting ground water, subsurface distribution of groundwater, geological formations as aquifers, types of aquifers, Groundwater flow, Darcy's law, Well losses, Well interference, Groundwater recharge, discharge and balance, Methods of Groundwater exploration

Hydrological Environments: Hydrogeological environments of Bangladesh, Sea-water intrusion in coastal aquifers, different types of aquifer in Bangladesh and its problems,.

Mass transport in Aquifers: Advection, Dispersion, Diffusion, Sorption, Retardation, Biodegradation.

Recommended References:

- Todd, D. K., and Mays, L. W. (2005), Groundwater hydrology Ed (Vol. 1625), Wiley, New Jersey. ISBN-13: 978-8126530038.
- Chow, V.T. Handbook of Applied Hydrogeology.
- Gupta, B.L. Water Resources Engineering and Hydrology.
- Kazmann, R.G. Modern Hydrology.
- Meinzer, O.E. Hydrology.
- Shaw, E.M. Hydrology in Practice.
- Todd, D.K. Ground Water Hydrology. Wisler, C.O. and Brater, E.F. Hydrology.

ESE 2103. Oceanography and Limnology

Credits: 2; Contact Hours: 2 Hours/Week

Oceanography

Marine ecosystems: Introduction, features of the oceans; Classification of marine environment (habitat).

Physico-chemical characteristics of marine environment: surface currents, waves, tides, continuity, light, temperature, pressure, viscosity and surface tension, density-temperature relationship, heat capacity, dissolved salts, major and minor ions in sea water, salinity-temperature relationship, dissolved gases and acid/base buffering.

Biota of marine ecosystem and their adaptations: General natures of marine life, Classification of biota of marine ecosystem - marine producers, shore communities, coral reefs - reef forming corals, zonation on coral reefs, symbiotic relationships in coral reef fish, coloration in coral reef fish, classification of marine plankton, migration of plankton, salinity effects on life, temperature effects.

Bay of Bengal: Morphologic features; Bengal Deep-Sea Fan sedimentation; Ecstasy and Relative Sea-Level Changes.

Principles of Palynology and Palynostratigraphy: Classification and distribution of Plants; Palynology of Spores, Pollens, Dinoflagellates etc. and their Paleoecology, Paleoclimatology and Paleoenvironments; Palynology of Bengal Basin and surrounding areas.

Limnology

Introduction: Definition, history and scope of limnology.

Physical features of inland water environments: Density, viscosity, surface film, surface tension, surface radiation, turbidity, effect of light and temperature on the growth of aquatic organisms, thermal stratification and circulation.

Productivity of aquatic habitat: Primary productivity and measurement of primary productivity in aquatic habitats; factors affecting primary production.

Wetlands of Bangladesh (Haors, Baors, Beels and Lakes of Bangladesh) Aquatic environments: classifications, definition of different types of water bodies, freshwater resources of Bangladesh.

Classification of lakes and special type of world lakes; Factors affecting growth and distribution of planktonic organisms.

Recommended References:

- Brasier, M.: Microfossils
- Cushman, A.J. Foraminifera
- Pokorny, V.: Principles of Zoological micropaleontology
- Sumich, J.L. 1992. In Introduction to the Biology of Marine Life, Wm. C. Brown Publishers, 3rd edition.
- Ragothaman, G. 2002. Aquatic Ecology, Trivedy, Agrobios, India.
- Subrahmanyam, N.S. and Sambamurty, A.V.S.S. 2000. Ecology, Narasa Publishing House, NewDelhi, India.
- Steve, N. Oceanography, 1987. John Wiley and Sons.

- Paul, S.W. 1952. Limnology (2nd ed.), McGraw Hill Book Company, USA.
- Ruttner, F. 1975. Fundamentals of Limnology (3rd ed.) University of Toronto Press, Toronto and Buffalo.

ESE 2105. Ecology

Credits: 2; Contact Hours: 2 Hours/Week

Introduction: Concept, definition, branches and scope of ecology;

Ecosystem- Concept and principles of ecosystem; structure, examples and function of ecosystem, Different types of ecosystem; Ecological tools and techniques; Statistical application in ecology;

Energy flow in Ecosystem: Energy in ecosystem, Pattern of flow of energy through the ecosystem and food chains; Biogeochemical Cycles of Ecosystem: Carbon cycle, nitrogen cycle, sulphur cycle and phosphorous cycle; Abiotic and Biotic component of Ecosystem; Phytosociology, Population ecology, Applied ecology.

Ecological factors and their impacts: Introduction, Climatic factors: Light, Temperature, Precipitation, Humidity, Wind-speed and their effects on vegetation; Edaphic factors: Soil types, profiles, components, factors affecting vegetation, soil- water, soil micro-organism and their roles, soil physical and chemical properties; Physiographic factors: Topographic conditions and their effects on vegetation, and Biotic factors: Types, interactions between plants, animals and man, interactions between plants and plants, and interactions between plants and micro-organisms.

Recommended References:

- Odum, E.P. ECOLOGY. Modern Biology Series. Oxford & IBH Publication Co.Ltd. New Delhi.
- Gopal, B. & Bhardwaj, N. ELEMENTS OF ECOLOGY. Vikas Publication Pvt. Ltd. India.
- Shukla, R.S.& Chandel, P.S. 1988. PLANT ECOLOGY AND SOIL SCIENCE. S. Chand & Co. Pvt. Ltd. Ram Nagar, New Delhi, India.
- Verma, P.S.& Agarwal, V.K. 1986. PRINCIPLES OF ECOLOGY.S. Chand & Co. Pvt. Ltd. Ram nagar, New Delhi, India.
- Edward J. Kormondy. CONCEPTS OF ECOLOGY. Prentice-Hall of India Private Limited. New Delhi-110 001.
- Biswarup Mukherjee. Environmental Biology.
- William G. Hopkins. 2002. Introduction to Plant Physiology, 2nd edition, John Wiley and Sons. Inc. New York.

ESE1202. Ecology –Lab

Credits: 1.5; Contact Hours: 3 Hours/Week

Introducing different tools and techniques of population and community study. Identification and preservation of different ecological samples; Field Survey- Determination of productivity of aquatic ecosystem.

Quantification of bloom forming plankton (phytoplankton + zooplankton); Determination of the biomass of producers in the given area; Study the biotic component of a pond ecosystem of JKKNIU Campus.

ESE 2107. Environmental Microbiology

Credits: 2; Contact Hours: 2 Hours/Week

Introduction to Environmental Microbiology: Introduction, Historical perspective, Branches of Environmental Microbiology, Scope of Environmental Microbiology, Modern Environmental Microbiology.

Aquatic Microbiology: Distribution of microorganism in fresh water environments, estuaries and oceans. Domestic and drinking water: pollution, purification, microorganism as indicator of water quality, bacteriological examination of water. Wastewater: characteristics and treatment processes.

Air Microbiology: Microorganisms in air, survival of microorganisms in air, techniques of the study of microorganisms in air, origin of microorganisms in air, control of microorganisms in air.

Soil Microbiology: Porous media, Characteristics of microorganism in porous media, Microbial activities in porous media, Microorganisms in surface soils, Methods of studying soil microorganisms.

Microorganisms in extreme environments: Psychrophiles, Thermophiles, Acidophiles, Alkalophiles, Halophiles and biotechnological applications of Extremophiles and Extremozymes.

Role of microorganisms in biogeochemical cycles: carbon cycle, nitrogen cycle, sulfur cycle and phosphorus cycle. Role of microorganisms in the degradation of hydrocarbons, other organic pollutants and remediation of heavy metals.

Interaction among microorganisms: Mutualism, Commensalism, Antagonism, Competition, Parasitism, Predation.

Recommended References:

- Microbiology: An Introduction. 8th ed. G.J. Tortora, B.R. Funke and C.L. Case. Pearson Education Inc, Singapore, 2007.
- Microbiology. 5th ed. M.J. Pelczar Jr., E.C.S. Chan and N.R. Krieg. Tata McGraw-Hill Publishing Co.Ltd., New Delhi, India,
- Manual of Environmental Microbiology, 2nd ed., C.J. Hurst *et. al.*. ASM Press, Washinton Dc, USA, 2002.
- Environmental Microbiology. R.M. Maier, I.L. Pepper and C.P. Gerba. Academic Press, Elsevier, 2006.
- Microbial Ecology- Fundamentals and Applications, 4th ed., R.M. Atlas and R. Bartha. Benjamin/ Cummings Publishing Co., Inc., 2000.
- Introduction to Soil Microbiology. Martin Alexander. John Wiley & Sons, Inc., 1961
- Soil Microbiology and Biochemistry, E.A. Paul and F.E. Clark, Academic Press, Inc., 1989
- Wastewater Microbiology. Gabriel Bitton. Wiley- Liss, Inc., USA., 1994
- Microbes in Extreme Environments. R. A. Herbert and G. A. Codd, Academic Press, Inc. 1986

ESE 2104. Environmental Microbiology- Lab**Credits: 1.5; Contact Hours: 3 Hours/Week**

Basic Rules and Requirements of a Microbiology Laboratory; Preparation of different types of culture media; Technique of serial dilution and isolation; Water quality test through faecal coliform counts by streak plate method / pour plate method / spread plate method and enumeration of coliform bacteria; Study of root nodules in legume plants. Sampling of bacteria from air, water and soil; Identification and Culture of microorganisms; Quantitative and qualitative study microorganisms from water, Soil, Fish, Shrimp and other fisheries products and growth study (factors affecting the growth of microorganisms, Temperature, Salinity, Osmotic pressure, pH)

ESE 2109. Survey and Settlement**Credits: 2; Contact Hours: 2 Hours/Week**

Types and Instruments of surveying. **Chain survey:** Plotting details, Errors, Merits and demerits. **Traverse survey:** Procedures, Bearing measure and plotting, Errors and adjustment, Merits and demerits, **Plane Table survey:** Orientation, Merits and demerits, Two and three point problems. **Leveling:** Objectives, Methods, Reduced level, Effects of curvature and refraction and their corrections. **Contouring:** Characteristics and uses, Direct and indirect methods. **Cadastral survey:** instruments, procedures and land settlement and map preparation. **Curves:** Types, Curve components, Setting of circular, transition and vertical curves, Super elevation and Shift. **Land survey:** Land ownership, property rights, laws governing property and definitions of land ownership, mapping and GIS (geographic information systems) software, GPS (global positioning system) handhelds and computer aided design. Analysis of highways and infrastructure. **Project survey:** Classification, Irrigation and Canal projects.

Recommended References:

- Punmia, B. C. (2005), Surveying (Vol. 1), Firewall Media. ISBN-13: 978-81-7008-079-4
- Aziz, M.A. and Shajahan, M. A text book of surveying
- Roy, S. K. (2010), Fundamentals of surveying. 2nd Ed., PHI Learning Pvt. Ltd. ISBN-13: 978-81-203-4198-2

ESE 2106. Survey and Settlement- Lab**Credits: 1.5; Contact Hours: 3 Hours/Week**

Chain survey, Plane table survey, Traverse survey, Leveling and Contouring, Route project, Tacheometry and Stadia surveying, Height and Distance problem, House setting, Curves and Curve settings. Use of Total Station and Global Position Station (GPS).

Recommended References:

- Punmia, B.C. (2005), Surveying (Vol. 1), Firewall Media. ISBN-13: 978-81-7008-079-4
- Aziz, M.A. and Shajahan, M. A text book of surveying
- Roy, S.K. (2010), Fundamentals of surveying. 2nd Ed., PHI Learning Pvt. Ltd. ISBN-13: 978-81-203-4198-2

ESE 2111: Soil Mechanics**Credits: 2; Contact Hours: 2 Hours/Week**

Definition of soil, volume composition of soil: Soil as a component of ecosystem; Minerals of soils, Importance of soil. Soil as a disperse three phase system, Mass volume relationship of soil constituents.

Soil structure and Morphology: Mechanism of aggregate formation and dispersion; classification on the basis of shape; importance of soil structure in relation of air-water content; Soil densities and pore space.

Stress and strain processes: Stress-strain relations and failure of soil bodies, Soil strength and its measurement, Soil compression, Soil compaction in relation to wetness, shear stress, stability of slopes in soils.

Soil water; Concepts of soil water, Factors affecting soil water retention, Methods of measuring soil water-gravimetric, tensiometer and neutron scattering method, Soil water potential-metric potential, pressure potential, gravitational potential, osmotic potential, Measurement of metric potential by tensiometer method

Water, contaminant and gas movement in soil: Infiltration-definition, factors affecting, equations, methods of measurement, Hydraulic conductivity-definition, Darcy's law, methods of measurement, classes of hydraulic conductivity, contaminant mass flux, movement of exchanges of gases in soil, solute transport in soil.

Soil temperature: Importance, factor affecting and management of soil temperature, thermal properties of soils, heat flow equation.

Mechanical analysis: Principle and methods, Derivation of Stoke's law.

Soil consistency and tillage: Forms of soil consistency, swelling, shrinkage, Tillage and puddling, Plough pan-formation, effects on soils and crops.

Groundwater hydrology: Groundwater flow, vertical flow of water, pollutant flow through groundwater, diffuse and point pollution, groundwater drainage and pollution, flow of confined and unconfined aquifers,

Recommended References:

- Daniel Hillel, 1998. Environmental Soil Physics
- Keller, E.A., 1992, Environmental Geology 6th ed.
- Graig, R. F., 2004: Graig's Soil Mechanics. London
- Punmia, B. C., 1994: Soil Mechanics and Foundations. Laxmi Publications Pvt. Ltd., New Delhi.
- Harrison, J. P and Hudson, J. A.: Engineering Rock Mechanics Part II

- Skinner, B.J. & Porter, S.C., 1995, The Dynamic Earth, 3rd edition; John Wiley & Sons, Inc. New York.
- Wild, A., 1996, Soils and the Environment, Cambridge Univ. Press.
- Miller, R.W. & Donahue, R.L., 1997, Soils in our Environment, Prentice Hall Private Limited.
- The Nature and Properties of Soils. Brady and Weil, 13th edition.
- Fundamentals of Soil Science, H.D. Foth, 6th edition, John Wiley and Sons.
- Fundamentals of Soil Physics, D. Hillel, 1st edition Academic Press, USA.

ESE 2108: Engineering Drawing-Lab

Credits: 2.0; Contact Hours: 3 Hours/Week

Fundamental Concepts: Drawing instruments and their uses; Lettering, numbering and constructing title strip, adopting symbols; Views; Projections: First angle, Third angle; Generation of views of solid bodies in different planes, Sectional views, Auxiliary views, Isometric views, Dimensioning, Basic concept of working drawing.

AutoCAD: Importance to design and drafting, Setting up a drawing: starting AutoCAD, menu, planning for a drawing, basic commands, making a simple 2-D drawing, layers, object snap, poly lines and other features, file handling and display control, editing and dimensioning.

Recommended References:

- I H Morris: Geometrical Drawing
- Hemanta Kumar Bhattacharia: Pratham Engineering Drawing
- R.W.P Aroogaswamy: Mining Geology

ESE 2113. Statistics for Environment

Credits: 3; Contact Hours: 3 Hours/Week

Frequency distribution: Measures of central tendency, dispersion; moments, skewness and Kurtosis. **Correlation and Regression:** Bivariate data, correlation and regression coefficients, regression. Probability: Sample space event, probability of events, theorem, of total and compound probability, conditional probability, bayes theorem. **Random variable:** Probability function, distribution function, joint, marginal and conditional probability function, expectation and moment generating function. **Parent Distributions:** Binomial, poisson, negative binomial, normal and exponential distribution. **Sampling Distributions:** χ^2 , t and F distributions. **Test of Hypothesis:** Test for population, mean, variance, correlation coefficient and regression coefficients.

Recommended References:

- Spiegel M R, Theory and Problems of Statistics
- Hoel P G, Elementary Statistics
- Hogg R V & Craig A T, Introduction to Mathematical Statistics, 4th Ed, Macmillan, London
- Mood, Graybill & Boes, Introduction to the Theory of Statistics, 3rd Ed, McGraw Hill, NY
- Mostafa M G, Methods of Statistics
- Milnyk M, Principles of Applied Statistics
- Barlow R G, Statistics
- Loveday, Practical Statistics and Probability
- Mosteller, Rourke & Thomas, Probability with Statistical Applications, 2nd Ed, Addison-Wesley, USA
- Goon A M & Gupta M N, Fundamentals of Statistics, Vol-I

2nd Year 2nd Semester

ESE2201. Forest and Wildlife Management

Credits: 3; Contact Hours: 3 Hours/Week

Basic Concepts of Forest and Wildlife Management: Forest and wild life as resource; Scope, objectives and importance of forest and wild life management; its development and status in Bangladesh.

Production of Timber from Forest: Factors of forest management (Land classification, Prescriptions and Yield predictions); growth, yield, and stand structure- concepts for forest management; timber production (site and density); models for predicting growth and yield.

Decision Analysis in Forest Management: A framework for decision making; financial analysis and the arithmetic of interest; evaluation of project and planning alternatives.

Forest Resource Management: Forest policy and law of Bangladesh and neighboring countries, History of forest resource management in Bangladesh, Current forest management regime, Institutional framework for forest resource management in Bangladesh, Constraints and scopes in forest resource management.

Park Management: History, concept, objectives, Size and legal status of national park; protected areas; park management programs and techniques; administration and maintenance; facility improvement and development; buffer zone; park protection and problems; economic value of park

Wildlife Conservation and Management: Principle and policy; Wild life conservation and management in Bangladesh and neighboring countries; Wild life conservation regulation (Act, order and ordinance), National and International organizations and their role in wildlife conservation.

Wetland Forest: Definition and scope, The Sundarbans mangrove swamp forest

Economic Value of Wildlife: Commercial value, Recreational value, Educational and scientific value.

Recommended References

- Ahmed, A. I. Deforestation and Forest Degradation in Bangladesh.
- Bailey, J. A. 1984. Principles of Wildlife Management. John Wiley and Son. N.Y.
- Balfour, E. 1985. The Encyclopedia of India. Vol. 1. London; Bernard Quaritch.
- Brockman, C. F. 1959. Recreational Use of Wild Land. McGraw-Hill Book Company.
- Davis, L. S. and K. N. Johnson. 1986. Forest Management: Decision Making Principles and Cases. W. B. Saunders Company.
- Duerr, W. A. 1979. Forest Resource Management: Decision Making Principles and Cases. W. B. Saunders's Company.
- Eisenberg, S. K. 1982. Elephant Blandford Press, Poole.
- Pant, M. M. 1990. Forest Management. Institute of Forestry, Chittagong University.
- SHED. Bangladesh Environment: Facing 21st Century.

ESE 2203. Environmental Geochemistry

Credits: 2; Contact Hours: 2 Hours/Week

Introduction: Definition, scope and goal of environmental geochemistry, its relation to Environmental sciences

Nature and general properties of solid phase of the soil: Clay minerals, oxides of metals and amorphous materials; structure and colloidal properties; soil organic matters-composition; humus and its classification according to origin; properties and influence of humus on soil properties and aquatic environment.

Aqueous Geochemistry: Natural/Fresh and Sea Waters and Redox processes.

Heavy Metals: Concept of major, trace and Rare Earth elements (REE) elements. Classification of trace elements, trace elements and health. Possible impacts of imbalance of some trace elements.

Groundwater Geochemistry: Solutions, Minerals and Equilibria; Carbonates and Carbon Dioxide; Ion Exchange and Sorption; Silicate Weathering; Saltwater and Mixing; Redox Processes. Eh-pH Relationships;

Geochemistry of Sediments and Rock: Geochemical Cycles; Rock-Water Interaction, Causes of soil acidification and alkalization, soil acidification, Influence of pH on solubilization of soil constituents and its environmental consequence. Geochemistry of clay minerals, Transformations of N, S, Fe, Mn and organic matter and their impact on environment, determination total organic carbon (TOC) and total nitrogen (TN) in soil and sediment, Bio-geochemical cycles (N, C, S and P).

Isotope Geochemistry and its Application: Environmental Isotopes: ²H, ¹⁸O, ¹³C, ¹⁴C, ¹⁵N, ³²S; Dating of Modern and Old Groundwater

Recommended References:

- N. C. Brady and R.R. Weil. The Nature and Properties of Soils, 13th edition, Pearson Education Asia.
- Kim H.Tan. Principles of Soil Chemistry, 2nd edition, Marcel Dekker Inc. New York.
- Kim H.Tan. Environmental Soil Science, 1st edition, Marcel.
- F.E.Bear (ed.). Chemistry of the Soil 2nd edition, Oxford and IBH Publishing Co. Pvt. Ltd.
- H.B.Mc. Bohn, Mc. Neil and G. O'Connor. Soil Chemistry, John Wiley and Sons, 1st edition.

M.B. McBride. Environmental Chemistry of Soils,
 S.E. Manahan. Environmental Chemistry, 7th edition, 2000.
 F.J. Stevenson. Cycles of Soil, 1st edition, John Wiley and Sons.
 F.N. Ponnampetuma. The Chemistry of Submerged Soils, In: Advances in Agronomy, 1992

ESE 2202. Environmental Geochemistry-Lab

Credits: 1.5; Contact Hours: 3 Hours/Week

Sampling and Analysis: Field instrumentation of hydrogeochemistry, procedures and design of soil/water sampling techniques; Techniques for Water and soil/sediment Analysis.

Water analysis: Laboratory instrumentation in water analysis; analytical methods; analysis of major anions and cations; analysis of trace elements; QA/QC procedures.

Water chemistry and data interpretation: Analytical accuracy check; source rock deduction; graphical interpretation; statistical analysis; spatial analysis; water quality parameters.

Digestion of soil and plants sample for metals: Plant and soil sample preparation, Solid sample processing procedure for digestion, Solid sample digestion procedure for metal analysis, Nitric acid ($\text{HNO}_3\text{-H}_2\text{O}_2$) and hydrogen peroxide digestion for metal analysis.

Synthetic Precipitation Leaching Procedure (SPLP), EPA Method: Preparation of extraction fluid acid mixture, Sample preparation, Leaching test (SPLP), Toxicity Characteristic Leaching Procedure (TCLP), California Waste Extraction Test (Cal-WET)

ESE 2205. Meteorology and Climatic Processes

Credits: 2; Contact Hours: 2 Hours/Week

Meteorology: Definition, Scope and Sub-classifications, History of meteorology, Nature, origin and composition of the atmosphere, Vertical divisions of the atmosphere; Meteorological equipment and Weather forecasting: Surface measurements, Application of remote sensing, Satellite observation.

Heating Earth's Surface and Atmosphere: Solar Insolation of the earth and heat budget, Variability of insolation, World distribution of insolation, Vertical distribution of temperature

Water balance of the atmosphere: Humidity, Physical changes of state of water Processes of cooling to produce condensation and sublimation, Clouds formation and classification, Fog formation and type, Precipitation, causes, forms and types.

Air Pressure and Winds: Pressure, vertical distribution of pressure, Pressure gradient and wind, Factors affecting wind speed and direction; General atmospheric circulation, Seasonal changes in the general circulation, Oceanic circulation, Monsoon, local winds, mountain and valley breeze; Stability and instability, air masses sources region and classification, Extra tropical cyclone, anti cyclone, Thunderstorms.

Climate controls and Climate: Definition, production of climates, Climatic classification of the world and World's climatic regions, the role of oceans, *El Nino*-Southern Oscillation (ENSO) events, *La Nina* events, preliminary concepts of climate change.

Recommended References:

- Franklyn W. Cole, Introduction to Meteorology, 3rd Edition, John Wiley and Sons.
- Roger G. Barry and R. J. Chorley. Atmosphere, weather and climate, 7th Edition,
- Edward Bryant, Climate Process and Change, Cambridge University Press, 1997.
- Trewartha and Horn, An Introduction to Climate, 5th Edition.
- IPCC report (recent edition)

ESE2204: Meteorology and Climatic Processes – Lab

Credits: 1.5; Contact Hours: 3 Hours/Week

Identification and application of meteorological instruments: meteorological data analysis

ESE 2207. Coastal Environment and Blue Economy**Credits: 3; Contact Hours: 3 Hours/Week**

Introduction: definition, importance, classification of coastal system; estuaries, tidal wetland, coral reefs, beaches and shoreline, erosion, accretion and coastal anthropogenic and natural disturbance of coast, Impact of coastal aquaculture on environment (e.g. shrimp culture) coastal resource evaluation of vital area, coastal plantation and coastal environment.

Coast protection: Multistructure embankment groans etc. their problems and remedies, establishment of coastal green belt, Development options in the coast, tourism and recreation, fisheries resource.

Coastal hazards and disaster management: Definition of hazards and disaster, difference among hazard, disaster and risk. Types and causes of coastal hazard.

Integrated Coastal Zone Management (ICZM): The process and need for ICZM, Coastal zone management issues, Integration of Agriculture, forestry and fisheries into ICZM, conflict resolution in ICZM.

Characteristics of Bangladesh coastal zone: Topographical changes, climate of coastal zone, soils of coastal zones and floodplain, Coastal embankment project, sea level rise and its implication for Bangladesh.

Blue Economy: Definition, Strategy, Oceans, Fisheries and Coastal Economies, The Transition to a Blue Economy, Challenges to the Blue Economy, Sectors of the Blue Economy-Fisheries, Aquaculture-Coastal and Maritime Tourism-Marine Biotechnology and Bioprospecting-Extractive Industries: Non-Living resources -Activities within National Jurisdictions -Activities in Areas Beyond National Jurisdiction-Desalination (freshwater generation) -Renewable Marine (off-shore) Energy-Maritime Transport, Ports and Related Services, Shipping, and Shipbuilding -Waste Disposal Management, The Potential of the Blue Economy: Increasing Long-term Benefits of the Sustainable Use of Marine Resources for Small Island Developing States and Coastal Least Developed Countries

Recommended References:

- Coastal environmental management plan for Bangladesh, Volume 2 Final report, Economic social commission for Asia and the Pacific, Bangkok, Thailand.
- PJS Boaden and R Seed (1985) An introduction to coastal ecology
- HC. Pereira (1989) Policy & Practice in the management of Tropical water shed, Balhaven press, London.
- FR Troch (1991) Soil and Water conservation, Prentice Hall. NJ.
- <http://hdl.handle.net/10986/26843>

ESE 2209. Environmental Biochemistry**Credits: 2; Contact Hours: 2 Hours/Week**

Introduction: Definition, the composition of living matter. Carbohydrates and their importance, oxidation of carbohydrates; Protein and its importance, amino acids, protein configuration, biological roles of protein;

Simple lipids: Fats and Oils, waxes, some physical and chemical properties of fats and oils; Nucleic Acids: Definition and Components, Phosphoric acid, Pentose sugar and Nitrogenous Bases. Nucleosides and nucleotides, DNA, Double helical structure, Denaturation, Renaturation and ribonucleic acid.

Enzymes: Importance, biological roles of enzymes, characteristics of enzyme, active site of enzyme.

Vitamins: General characteristics of vitamins, fat-soluble vitamins, water-soluble vitamins; their occurrences, deficiency and human requirements.

Metabolism: catabolic pathways, gluconeogenesis, energy production in aerobic processes, anaerobic metabolism, biosynthesis, control of metabolic processes.

Recommended References:

- Albert L. Lehninger. Principles of Biochemistry. 2nd Ed. CBS Publishers & Distributors. New Delhi, India, 1993.
- EDP De Roberties and EMF De Roberties, Cell and Molecular Biology. Jr. Lippincott Williams & Wilkins, 2001.
- N. Rajvaidya and O.K. Markandey. Advances in Environmental Sciences and Technology, Vol. 3: Environmental Biochemistry. A.P.H. Publishing Corporation, New Delhi, India. 1998
- C. Ratledge and B. Kristiansen. Basic Biotechnology, Part I: Fundamentals and Principles. 2nd ed. Cambridge University Press, 2001.
- S. E. Manahan. Fundamentals of Environmental Chemistry, Lewis Publishers, 2000.

ESE 2211. Biodiversity Conservation**Credits: 2; Contact Hours: 2 Hours/Week**

Biodiversity: Introduction (Definition and concepts, Genetic, species and ecosystem biodiversity), Measuring biodiversity (Number and differences, surrogacy, Richness, Evenness), the value of biodiversity (Ecological value, Ethical value), threats of biodiversity (Extinctions, extinction of past, human caused extinction)

Conservation: Definition and concepts, importance of conservation, the rise of modern conservation, Approaches of Conservation, biodiversity conservation inside protected areas (establishing and designing and measuring protected area), biodiversity conservation outside protected area, International Agreement on Biodiversity conservation.

Threats and Conservation of Biodiversity: Depletion of Biodiversity, Causes of threatening Biodiversity, IUCN categories of endangered species, Red data book, Measures for Biodiversity conservation, protected area and biosphere serves Role of Zoos, Botanical gardens and captive breeding programme.

Recommended References:

- Odum, E.P. ECOLOGY. Modern Biology Series. Oxford & IBH Publication Co.Ltd. New Delhi.
- Gopal, B. & Bhardwaj, N. ELEMENTS OF ECOLOGY. Vikas Publication Pvt. Ltd. India.
- Shukla, R.S.& Chandel, P.S. 1988. PLANT ECOLOGY AND SOIL SCIENCE. S. Chand & Co. Pvt. Ltd. Ram Nagar, New Delhi, India.
- Verma, P.S.& Agarwal, V.K. 1986. PRINCIPLES OF ECOLOGY.S. Chand & Co. Pvt. Ltd. Ram nagar, New Delhi, India.
- Edward J. Kormondy. CONCEPTS OF ECOLOGY. Prentice-Hall of India Private Limited. New Delhi-110 001.
- Biswarup Mukherjee. Environmental Biology.
- William G. Hopkins. 2002. Introduction to Plant Physiology, 2nd edition, John Wiley and Sons. Inc. New York.

ESE 2206. Field Work**Credit: 1.0**

To be introduced different types of environment, resources and natural diversity in Bangladesh through fieldwork at different places and/or related organizations. Each student shall require submitting a report and presenting a seminar group-wise according to fieldwork.

ESE 2208. Viva Voce**Credit: 1.0**

Comprehensive oral examination of the courses taught.

ESE 3101. Environmental Pollution

Credits: 3; Contact Hours: 3 Hours/Week

Introduction-Definition of Environmental Pollution; Types of Pollution, Pollutants and Contaminants; Natural and Anthropogenic Pollution

General Concepts of Pollutants: Energy –nuclear, solar, fossil; Transpiration: - Rail, road, air, agricultural activities, fertilizers, pesticides, medicine, drugs, sedative drugs, herbal drugs, detergents, polymers, dyes.

Air pollution: Fundamentals of air pollution problems- Ambient Air: Sources & types of air pollution; Air pollution by automobiles; Adverse effects of air pollution and pollutants on inert materials, vegetation, man and animals; Air pollution control; Classical Smog and photochemical smog; Smoke; Gaseous contaminants: Dust, fumes, and mists; Sampling and monitoring of air pollutants; Measurement methods: Laws and regulation of air pollution control;

Water Pollution-Definition, major sources of water pollution, Nature and types of water pollutants: Heavy metals, organic-inorganic substances, pesticides, PCB and Radionuclides, Effects of water .

Noise pollution: Effects, sources & control of noise pollution;

Thermal pollution: Sources, effects and control of thermal pollution; Thermal pollution over different location; Possible long-range impact of thermal pollution;

Radioactive Pollution-Definition, different sources of radioactive pollution: Natural and manmade sources, Applications and effects of radioactive materials on human and environment, Nuclear accidents, Biological magnification

Soil Pollution: Effects of pH in soils; Soil reactions and correlation; Tolerance of higher plants to saline and sodic soil; Soil erosion, erodability factors, topographic factors; Computation soil losses; Effects of pesticides on soil and soil organisms; reaction of inorganic contaminants in soil; Soil pollution and its impact on the environment, ground water,; Soil improvement methods for removal of pollutants.

Recommended References:

- Dara, S.S. (2002). Environmental chemistry and Pollution Control. S. Chand & Company Ltd.
- Bhata, S. C. Environmental Pollution and Control in Chemical Process Industries. Khanna Publishers.
- Sharma, B.K. Environmental Chemistry.
- Botkin and Keller. Environmental Science.
- De, A. K. Environmental Chemistry (6th Edition). New Age International Limited Publishers.
- Miller, G. T. Introduction to Environmental Science.

ESE 3102. Environmental Pollution-Lab

Credits: 1.5; Contact Hours: 3 Hours/Week

Detection of suspended particulate matter (SPM), SO_x, NO_x, CO₂, CO in air. Chemical analysis of water and wastewater, Measurement of sound levels. Air sample test (SO_x, CO, NO_x, particulates, Hydrocarbons, Lead determination); Determination of Pesticide residue; Measurement of intensity of noise, Measurement of temp. Relative humidity, Observation of daily surface temp. of various media, Measurement of moisture content of soil, Measurement of wind speed etc. and Based on Theory Classes of Environmental Pollution Control Engineering.

Soils: Concepts of soil, major components of soil, chemical and mineralogical composition of the earth's crust, physical, chemical and biological properties of soil.

Soil genesis: Rocks and minerals, weathering, soil forming factors, soil forming process, soil profile and horizon characteristics.

Soil classification: Soil classification, kinds and modern system of classification in the world, classification of Bangladesh soils and their correlation with international systems.

Soil physical properties: Soil texture; soil structure; soil consistence; density and weight relationships; soil pore space and porosity; soil air, soil color; soil temperature.

Soil taxonomy: Diagnostic surface horizons; diagnostic subsurface horizons; soil moisture regimes; categories of soil taxonomy; the abac soils; the pedon; classification of soils of Bangladesh and their correlation with other systems of classifications.

Soil water management: Soil water, soil water constants, soil water energy continuum; energy and pressure relationships; the soil water potential; soil water movement; plant and soil water relations; soil water regime; water conservation; soil drainage; irrigation.

Soil chemistry: Ion exchange; soil acidity, soil pH and nutrient availability, liming, soil buffering; base saturation percentage, mechanism of nutrient uptake by plants.

Soil erosion and conservation: Predicting erosion rates on agricultural land; application of the soil-loss equation; the soil loss tolerance value; water erosion on urban lands; wind erosion; factors affecting wind erosion.

Soil fertility management: organic matter depletion and its management; essential plant nutrients and their functions, plant–nutrient relationship; fertilizer; soil fertility evaluation and fertilizer management.

Recommended References:

- Henry D. Foth (1990). Fundamentals of Soil Science. 8th ed. John Wileys & Sons Inc. New York.
- Nyle C. Brady & Ray R. Weil (2008). The Nature and Properties of Soils. 14th ed. Prentice Hall, Upper Saddle River, New Jersey – 07458.
- Brammer H (1996). The Geography of the soils of Bangladesh. University Press Ltd, Dhaka, Bangladesh.

Pollution and human health: Trace element deficiency and disorders, occupational health hazards, biogeochemical factors in environmental health, epidemiological issues- goiter, fluorosis, arsenic poisoning.

Transmissible diseases: Symptoms, epidemiology and control of vector borne diseases-amoebiasis, trypanosomiasis, filariasis, leishmaniasis, schistosomiasis, life cycle of Plasmodium, control of Malaria, tuberculosis and AIDS. Waterborne diseases: Jaundice & diarrhea.

Occupational Health: Introduction, Occupational health hazards, Occupational diseases, Prevention of Occupational diseases.

Primary Health Care (PHC): Concept and scope, Health for all, Definition of PHC, Principles, Components, strategy of PHC, National Health Programme, Essential service package, National and International health organizations and regulations.

Environment, Health and Safety Measures: Concept of safety measures in work places, roles and responsibilities of employers and employees to ensure safety in work places, steps to protect health and safety, managing health and safety in work places, Hazard management at work places, occupational health and safety training at work places.

Recommended References:

- Jahan and Ahmed. Participatory Management of Low-coast Water Supply and Sanitation.
- Park, K. Textbook of Preventive and Social Medicine.
- Rashid, K. and Haider. Textbook of Community Medicine and Public Health.
- Russell, B. (2012). Work place Health and Safety Hand Book. SafeWork SA

Introduction to Energy and Environment: Definition, classification and sources of Energy, Importance and Energy use, Energy demand and energy supply, Transformation of primary energy to secondary energy, final energy, Automobile; major source of Environmental pollution

Fossil Fuel (Biomass, Oil, and Natural Gas): Source, Reserve of Fossil fuel, Definition and types of fuel, Calorific value of fuels, Types of coal and gaseous fuels, Advantages of liquid fuels over solid fuels, Advantages of gaseous fuels over solid fuel, Environmental Problem associated with Fossil fuel, Transboundary effect of fossil fuel burning

Nuclear Energy: Source and nuclear process; Fission, Fusion; Nuclear power plant; Environmental Problem associated with Nuclear Power Plant (Potential impact on environment); Waste disposal system of nuclear power plant; Safety of nuclear power plant; Advantages of nuclear power plant; Site selection of nuclear power plant

Solar Energy: Definition and classification of solar energy, solar energy storage; Solar energy collectors, solar cells, solar pumping, solar cooker, solar dryer; Advantages and disadvantages of solar energy

Geothermal Energy: Definition and sources of geothermal energy; Application or uses of geothermal Energy; Limitations and environmental consideration of geothermal energy

Hydro and Wind Power Energy: Definition of hydro, tidal and wind power energy; Hydroelectric power plant; Wind mill, site selection of a windmill, performance of wind machine; Main components of tidal power plants, classification, advantages and disadvantages of tidal power plant, site selection

Fuel Cell and Biogas Power Plants: Definition and operating procedure of fuel cell; Definition and composition of biogas; Classification and operation of bio-gas plant, advantages and disadvantages

Energy Conversion: Properties of fuels; Combustion processes (Complete, Incomplete, and Internal); Basic gas laws and combustion equations; Measurement of energy conservation; Energy efficient technologies

Energy Scenario of Bangladesh: Sources, reserve and uses of energy; Energy strategy of Bangladesh; Energy, environment and politics

Recommended References:

- Bala, B. K. 1997. Energy and Environment-Modeling and Simulation. NOVA Sci. Pub.
- Bala, B. K. Solar Drying Systems-Modeling and Simulation.
- Blahden, J. R. 1996. Energy Resources and Environment. Hadder and Stoughton, The Open University.
- Carla W. Montgomery. Environmental Geology (6th edition). McGraw-Hill.
- Chauhan, D. S. and S. K. Srivastava. Non-conventional Energy Resources.
- Godfrey Boyle ed. Renewable Energy 2nd edition, Oxford University Press.
- Nagpal, G. R. Power Plant Engineering.
- Sukatme. Solar Energy. Tata McGraw-Hill Publishing.

Introduction of Remote Sensing: Introduction and Scope of remote sensing, Definition, principles, spectral bands and structures, Scope of Remote sensing for environmental science

Satellites: History, types of satellites and their specialization, orbit characteristics, sensor characteristics, application

Radar: Remote sensing images, Preprocessing and processing techniques, enhancement techniques, filtering; Digital image classification.

Aerial Photograph: Introduction, Basic photo interpretation techniques, photo scale and stereoscopic parallax, color concept, flight planning.

Remote-sensing Data Processing and Application: Spectral Reflectance of Vegetation, Soil and Water, Application in geological studies- Floods and Water resources, Slope stability analysis, Displacement mapping of faulting, Earthquakes, Geobotany, etc.

Introduction of GIS: Basic concept, History of GIS, Definition, Scope; Fundamentals of GIS; Maps and Map Projections; Spatial Data Models; Raster Structures; Vector Data Structures; Data Sources; Data Quality; Spatial Data Input; Digitizing, Editing, Data Base, and Database Management; Relational Databases; Attribute Data; GIS Analyses; Application of GIS in various fields, Spatial analysis and modeling .

Recommended References:

- Bonham-Carter, G. F. 1994. Geographical Information System for Geoscientists-Modeling with GIS.
- Clarendon-DeMers, M. N. 1997. Fundamentals of Geographical Information Systems. Wiley.
- FAO, 1991. Use of High Resolution Satellite Data for coastal Fisheries. RSC Series no. 5., Remote Sensing Center.
- Star, J and Estes 1990. Geographic Information Systems- An Introduction.
- Joseph, G. 2005. Fundamentals of Remote Sensing (2nd ed.)
- Lillesand, T. M. and R. W. Kieffer. 1994. Remote Sensing and Image Interpretation (3rd ed.). John Wiley and Sons.
- Pergamon-Burrough, P. A. 1986. Principles of Geographical Information Systems for Land Resource Assessment. Schowengerdt, R. A. 1997. Remote Sensing: Models and Methods for Image Processing (2nd ed.). Academic Press. UN. 1997. Space application for Sustainable Development.

Remote Sensing Lab:

ERDAS Imagine and Ilwis based RS lab will include: Supervised classification of multispectral imagery such as Landsat, Aster and MODIS imagery; Feature space classification technique; Mapping of geologic structures (folds/faults) of Sylhet and CHT areas through digitization; creation of shapefiles; Change Detection; Post classification processing including accuracy assessment and map composition; Filtering of RADAR imagery; Simple classification of Radar images using thresholds; Unsupervised classification of Radar images;

GIS Lab:

ArcGIS based GIS lab will include: Project based lab work that combines the analysis techniques taught in theory course. The instructor will demonstrate the techniques using the software tutorial. Then the students will develop their projects and these will be approved by the instructor. Some of the data layers should be generated from image analysis done in the first half of the lab. Output of the project will be presented in class.

Recommended References:

- Bonham-Carter, G. F. 1994. Geographical Information System for Geoscientists-Modeling with GIS.
- Pergamon-Burrough, P. A. 1986. Principles of Geographical Information Systems for Land Resource Assessment. Clarendon.
- Clarke, K.C., 2003, Getting Started with Geographic Information System. 4th edn., 340p., Prentice Hall, New Jersey. Bonham-Carter, G.F., 1994, Geographic Information Systems for Geoscientists: Modeling with GIS. 398p. Elsevier Science Publications.
- Lillesand, T.M., Kiefer, R.W. and Chipman, J.W. 2004, Remote Sensing and Image Interpretation. 5th edn., 763p., John Wiley and Sons. ISBN 0471152277

ESE 3111. Solid Waste Management**Credits: 3; Contact Hours: 3 Hours/Week**

Solid Wastes: Introduction, classification, Problems and hazards of SWs, Management of Municipal SWs and Radioactive SWs

Landfill Leachate: Definition and types of hazardous waste, characteristics, hazardous waste combustion, Storage, treatment and disposal of hazardous waste, Control of major accident hazards, toxic substances and their removal technologies.

Hazardous and health-care waste management: Introduction, classification, Effects of hazardous wastes, Management of hazardous waste.

Radioactive waste : Definition and types of radioactive waste, radioactive waste management including local and centralized waste management, Radiation symbol and labeling, Legislative framework, Radiation poisoning, radiation monitoring, Nuclear and radiation accidents, Chernobyl accident, Nuclear weapon, trends and practices of radioactive waste management in Bangladesh.

Chemical Aspects of Wastes: Organic compounds in the environment (petroleum, Amino acids and proteins. Lipids, Carbohydrates, Humic substances, Lignins, etc.), Toxic chemicals in the environment, organic micro pollutants, Biologically essential elements, Organometallic compounds and their properties, Radionuclides and Radioactive Materials, Radioactive decay and Radioactivity, Radionuclides of anthropogenic origin.

Solid Waste Management Technology: Engineering design and operational aspects of waste generation, collection, storage, transfer, processing, including composting of organic waste, treatment and disposal; Engineering evaluation of: integrated waste management, solid waste characterization and classification, reduction, reuse and recycling, resource recovery and utilization; Life cycle assessment of waste, physical and chemical treatment methods and composting; Landfill design and operation including: site selection, engineered sites, liners and covers, leachate control and treatment, gas recovery and control, Planning and socio-economic aspects of solid waste management.

Recommended References:

- American Water Works Association. 1990. Water Quality and Treatment- A Handbook of Community.
- Arceivala, S. J. 2005. Wastewater Treatment for Pollution Control. Tata McGraw-Hill Publishing Company Ltd. New Delhi, India.
- Dara, S.S. (2002). Environmental chemistry and Pollution Control. S. Chand & Company Ltd.
- Eddy, M. 1996. Waste Water Engineering. McGraw-Hill Inc.
- Gurnham, C. F. Principles of Industrial Waste Treatment. John Wiley and Sons Inc., New York.
- Hammar, M. J. and M. J. Hammar. 1996. Water and waste water technology. Prentice Hall Inc.
- Pontians, F. W. Water Supplies. McGraw-Hill Inc.

Introduction: History and evolution of environmental laws, Protocol and ethics, Legal aspects of environmental resource management, Concepts and principles of environmental laws and its legislative

Mention the major UN conferences regarding environment: Stockholm declaration, 1972, Agenda 21, Rio-declaration, Kyoto protocol, CDM and Carbon credit, Convention of biological diversity, Ramser convention, Basel convention, UN convention on the laws of the sea; and other recent conventions.

Environmental law regime in Bangladesh: constitutional provision, role of BELA in the legal activism in Bangladesh, Bangladesh position on ICTPs; ECA 1995, ECR 1997, EC 2000 and other recent laws, Environmental policy: National and international.

Sectoral environmental legislation's in Bangladesh: laws regarding Biodiversity conservation, Forestry, Fishery, Mineral exploration, Groundwater management etc.

Relation of Environmental Laws to Ethics and ethical issues in environment. Role of classical and contemporary ethical theories to the suggestive measures for protecting environmental degradation both at local and global levels.

International Environmental Conventions: Protocols and treaties and their implementation in Bangladesh, International organization involved in environmental laws, protocols, conventions and treaties (UNEP, UNDP, IUCN, WWF, etc).

Comparative Study: Environmental laws in developed and developing countries and its enforcement, Environmental laws and policy in Bangladesh

Recommended References:

- Mohanty S.K., Universal's Environment & Pollution Law Manual, Universal Law publishing Co. Pvt. Ltd., 1999.
- Laws Regulating the Environment in Bangladesh, M. Farooque and R. Hassan, published by BELA.
- Rahman A.A., Haq S., Haither R., Jansen E.G. (eds), Environment & Development in Bangladesh, voll 1&2, UPL, 1994.
- Pojman P.L., Global Environmental Ethics, Mayfield Publishing Company, 2000
- Rolston-3 H., Environmental Ethics, Temple University press, Philadelphia, 1998.

Introduction to Natural Hazards: Importance; processes; Hazards, disaster or catastrophe; Death and damage.

Environmental hazards: A Conceptual Background; Fundamental Concepts for Understanding Natural Processes as Hazards; The Tool of Environmental Hazards; Environmental Hazards and Development; Perspectives on Prevention and Reconstruction of Natural Hazards; Reducing the Impact of Natural Hazards.

Hydrological Hazards (Floods, Tropical Cyclones): Flood: A Conceptual Clarification, Types of Flood, River Floods, Coastal Floods, Causes of Floods, Beneficent Effects of Floods, The Negative Effects of Flood ; Drought: A Conceptual Review, Desertification: A Conceptual Review, Indicators of Desertification, Causes of Desertification, Major Impact of Drought and Desertification , Socio-Economic Impacts, Control of Desertification.

Geological Hazards (Earthquakes, volcanic/eruptions, landslides): Introduction, Seismic Waves, Magnitude and Intensity, Occurrence and Severity of Earthquakes, Ground Motion, Ground Failure, Tsunamis and Coastal Effects, Fire, Volcanic Tectonic, Primary volcanic hazards, Ground deformation, Lateral Slides and Debris Avalanches, Environmental control, Methods of Diverting and Controlling Lava Flows, Conceptual Review of Landslide, Landslides Terrain, Landslides: Strength of the Material, Types of Landslides, Rotational slides, Translational slides, Causes of landslides, Rock Falls, Debris flows.

Coastal Hazard and Disaster Management: Types and causes of coastal hazard, Adjustment of hazard, Warning and forecasting system, case studies in Bangladesh.

Hazard assessment: technique, uncertainty, warning system, types and levels, natural resources maps, hazard related maps, use of hazard information in the development and planning process, stage of development planning, integrated flood hazard assessment in to development planning process.

Disaster and Hazard Reduction: Scope of mitigation of environmental disaster and hazards, methods of mitigation, livelihood adaptation, risks reduction model, institution (flood action program, FAP), early warning information and GIS based disaster technology.

The significance of disaster: The disaster threat; National Disaster management Policy; Major requirements for coping with disaster;

Recommended References:

- Smith, 1995, Environmental Hazard, 2nd edition.
- UNDRO, 1991. Mitigating Natural Disasters Phenomena, Effects & Options a manual for policy makers and planners, United Nations, New York.
- Lundgren L (1986) Environmental Geology. Prentice Hall
- Keller EA (1985) Environmental Geology. CBS Pub and Distributor, Delhi
- Edward A. Keller, 1985, 4th edition. Environmental Geology. CBS Publishers & Distributors, Delhi.
- R. Kerry Rowe .2001, Geotechnical and Geoenvironmental Engineering Hand Book. Kluwer Academic Publishers, London.
- Roy E. Hunt, Geotechnical Engineering analysis and Evaluation.
- Applied Geology for Engineers, by: Military Engineering volume XV.
- P. Leonard Capper, W. Fisher Cassie, and JD. Geddes, Problems in engineering soils,
- Roy E. Hunt, geotechnical engineering investigation manual.

Instruction: Definition, sources of wastes, Classification of wastes (solid waste, liquid waste, etc.), Characteristics of different types of wastes.

Liquid Wastes: Introduction, classification, Effects of liquid wastes, Management of liquid waste, Industrial effluent management, Sewage management, Clinical/hospital liquid waste management.

Wastewater type and generation: Characteristics, quality and quantity of municipal and industrial wastewater; Problems associated with municipal and industrial wastewater;

Basic overview of wastewater treatment: collection systems, collection system types, components and problems; preliminary treatment processes, process purpose and equipment, sedimentation tanks, settling tank effluent .

Objectives of wastewater treatment: Principles of physical, chemical and biological treatment processes; Preliminary or preparatory treatment: Screening, comminutors, grit removal, floatation, neutralization, equalization, aeration, skimming tank, for industrial wastewater- sour water strippers, heavy metal removal, air stripping, chemical oxidation;

Wastewater treatment and disposal methods of major industries - such as petroleum industries (gasoline kerosene treatment), textile industries, tannery; cement, fertilizer, paper and pulp, jute processing, dairy, drug and pharmaceutical, food and allied industry; Treatment and disposal of industrial waste sludge; Laws and regulations for industrial wastewater and waste treatment.

Biological treatment: process systems, trickling filters, rotating biological contactor (rbc), treatment ponds and lagoons, activated sludge systems, secondary settling.

Wastewater disinfection and discharge: chlorination, dechlorination, w (ultraviolet) radiation, ozonation, bromine, wastewater discharge, wastewater reuse.

Water solids management: alum sludge treatment processes, softening sludge (lime sludge) treatment processes, water sludge treatment processes, disposal and recoverable and non-recoverable water.

Wastewater reclamation and reuse: Methodologies, regulations of effluent disposal into lakes, reservoirs, rivers and ocean.

Design and Management of Effluent Treatment Plant (ETP)

Recommended References:

- Joanne E. Drinan, 2001 *water and wastewater treatment: a guide for the non-engineering professional*, CRC Press LLC
- Mara, d. Duncan, 2004, *Domestic wastewater treatment in developing countries*, Earthscan in the UK and USA.
- Hammer M.J. and Hammer M.J Jr. 1996. *Water and Waste Water Technology*, Printice Hall.
- Metcalf and Eddy. (1972), *Wastewater engineering: collection, treatment, disposal*. McGraw-Hill. ISBN-13: 978-0-07-041675-8
- Ahmed, M.F., and Rahman, M.M. (2000), *Water supply and sanitation: Rural and low income urban communities*. ITN- Bangladesh, Centre for Water Supply and Waste Management, BUET.

To be designed on the basis of theoretical classes.

Assessment of characteristics of wastes (Color, Odor, P^H BOD, COD, TSS, TDS, etc.), designing consideration of ETP.

Introduction to Environmental Sustainability, Importance, Fundamental Aspects of Environmental Sustainability; Overview of Major Environmental Issues, Global Environmental Issues. Economics and Sociology to Urgent Environmental Problems; Air, Water & Noise Quality Issues; Ecology, Natural Resources, Energy Cycles and Accounting; Carbon Cycle, Emissions and Sequestration; Concepts of Green Design; Green Chemistry; Life-cycle Analysis; Ecosystem degradation and resource limitations, Millennium Development Goals (MDGs); Economic Development vs Environmental Sustainability; Development and Displacement; Urban planning and Sustainability; Environmental governance/justice and resource conflicts; Environmental Sustainability in Bangladesh.

Recommended References:

- Allen, D.T., Shonnard, D.R, *Green Engineering: Environmentally Conscious Design of Chemical Processes*. Prentice Hall PTR 2002.
- Mukesh Doble and Anil Kumar Kruthiventi, *Green Chemistry and Engineering*, Elsevier, Burlington, USA, 2007.

Concepts and Systems of Environmental Economics : Definition and types of economic system, Price determination and resource allocation, Markets and regulatory approaches of national income, Environmental protection and its macro-economic consequences, Demand and Supply Curves.

Economy and Environment: Natural resource economics, The fundamental balance, The environment as an economic and social asset, The relationship between economic growth, population and environmental quality

Benefits and Costs of Environment: Cost-effectiveness analysis, Benefit-cost analysis, Environmental decision making- public policy development, Environmental risk analysis for environmental decision-making, Environment, economics and politics

Economics of Environmental Quality: Pollution control- a general model, Pollution damages, Abatement costs, The efficient level of emissions, Enforcement costs

Economics of Enforcement: Environmental taxes and green tax reform, Incentive-based strategies, Emission charges, Abatement subsidies

Institutional Strengthening and Capacity Building: Policies: liability laws, property rights and moral suasion, Command and control strategies, The economics of standards

Recommended References:

- Mankiew, N, G. (2014) Principles of Macroeconomics, 7th edition, South-Western College
- Mankiew, N, G. (2014) Principles of Microeconomics, 7th edition, South-Western College
- Parkin, Michale, (2003) Microeconomics, 6th Edition, Addison Wesley
- Samuelson and Nordhaus (2005), Economics, 18th Edition, McGraw Hill International.

Basic Concepts of Water Resource: Integrated Water Resource management (IWRM) concept, Characteristics of WRS, Planning of IWRM, Water policy, Allocation mechanisms of water, Water demand, Uncertainty and risks in IWRM, Evaluation of strategies of water management, Multi-criteria analysis.

Planning Fundamentals and Processes: Water resource systems, Sustainable development and uses of water, Agricultural perspectives and flood control system, Watershed management;

Water Supply: Importance, objectives and elements of water supply, Sources of supply and distribution system, Planning and design consideration;

Financial Aspects from Supply Side: Cost of water supply systems, Cost sharing, Financing of water supply systems, Willingness to pay;

Water conservation and demand management: Water conservation, Necessity of water conservation, Define demand management, Prerequisites for demand management, Tolls for demand management;

Bangladesh Perspectives: Water policy of Bangladesh, Ground water in Bangladesh, Environmental and Institutional aspects of water resource planning and management, d) Problems regarding water management in Bangladesh;

International Linkage: Water sharing treaty and analysis, Plausible impacts of Tipaimukh dam and Farakkah barrage in Bangladesh, By lateral and multilateral agreement in water sharing. Transboundary water resources and hydropolitics.

Coastal Engineering: Coastal structure, Impact on coastal zone and force on coastal structure; Beach and port management; Sediment properties; Transport process: Along shore line sediment transport process, wind transport of sand; Management of coastal environment: Coastal water quality, Coastal water supply and sewage, Intrusion of saline water to estuary; Coast pollution; Preservation of coastal environment.

Recommended References:

- M C Chaturvedi. Water Resources Systems Planning and Management. Tata McGraw-Hill Publishing Company Ltd. New Delhi, 1987.
- A MacDonald. Developing Groundwater A Guide for Rural Water Supply –J Davies, R Calow and J Chilton, ITDG Publishing, UK, 2005.
- Warren Viessman, Jr and Mark J Hammer. Water Supply and Pollution Control. Person Prentice Hall, New Jersey, 2005

ESE 3211. EIA and Environmental Auditing**Credits: 2; Contact Hours: 2 Hours/Week**

EIA - important terms in environmental impact assessment; Aims and objectives of EIA; Role of EIA in EM; EIA Methodology: Different methods of EIA, initial environmental examination, Baselines studies; Rapid environmental impact assessment methods: check list, matrix methods, Scooping and people's participation in EIA; EIA of development schemes; Economical evaluation of EIA; Cost and Cost/benefit analysis of EIA; EIA (State of art) in water resources and industrial projects; Application of EIA in energy and agricultural development; EIA for embankment flood protection measures; EIA of irrigation activities: EIA of infrastructure development project; Environmental impact of flood in rainy season; EIA of draughts in dry season (specially Farakkah and other upstream barrages); EIA in solid waste management; EIA of gas field and tea garden; EIA in JMB project; EIA of transport system; EIA procedure in private sector, EIA procedure in govt. sector; Different EIA index calculation.

Environment Risk Assessment (ERA): Definition, legal framework, risk evaluation, risk characterization, public perception of risk Major steps in ERA , Risk characterization and comparative risk assessment, risk analysis, industrial issues and health risk, case studies

Environmental Regulation and Auditing: Hazardous waste legislation, protocols, institutional and legal framework, ISO 14000, the nature of environmental auditing, audit planning, basic component of auditing, beneficiary groups.

Recommended References:

- P.Watheru, 1992, Environmental Impact Assessment – Theory and Practice, Chapman and Hall Ltd, New York.
- L.Q. Center, 1996, Environmental Impact Assessment, McGraw-Hill, Inc, New York. N.Harley and C.L.Spash, 1998.
- C.J.Barrow, 1997, Environmental and Social Impact Assessment, John Wiley & Sons, New York.

ESE 3213. Agriculture and Environment**Credits: 2; Contact Hours: 2 Hours/Week**

Historical perspective of agriculture: Environmental factors and adaptation of agricultural plants: Basic principles of agricultural crop production, Concept of agroforestry and management, Soil, land and cropping pattern in Bangladesh, agro-ecological zones in Bangladesh.

Agro-meteorology: Introduction, objectives, data sources, estimation crop yields, How to monitor and use meteorological data for agriculture purpose, speculation of future trend of agricultural changes.

Agrochemical and Environment: Historical perspectives of agrochemical uses, Basic principle of agriculture crop production, Natural hazards, types of agrochemicals, Effect of agrochemicals on soil, Fresh water and plant kingdom, Agrochemical residues, misuse of agrochemicals, Pollution Control in Agriculture: Slow release fertilizer, Integrated Pest Management (IPM).

Impact of climate changes on agricultural production: Bangladesh and global perspectives, understanding the driving forces of land use changes in Bangladesh, option for mitigation of the impact of climate changes, environmental changes and food security of Bangladesh and other densely populated countries.

Natural problems in agriculture practices: flood, drought, cyclone, rainfall, salinization, pest and disease, mitigation of natural problems.

Agricultural practice and technology: Choice of technology and technology transfer: import of environmental friendly technology from other Asian countries, Sustainable agricultural development in Bangladesh.

Production of transgenic plants: micropropagation, secondary metabolites production, impact of genetically modified food and biosafety.

Recommended References:

- Mishra SJ and Mani D (1994) Agricultural pollution. Ashish Pub, New Delhi.
- Whyte RO (1980) Crop Production and Environment. Faber and Faber Ltd, London.
- Hugh Brammer (1999) Agricultural disaster Management in Bangladesh, UPL, Dhaka.
- Chawla H.S. (2002) Introduction to plant biotechnology, 2nd Ed. Oxford & IBH publishing Company

ESE 3204. Field Work**Credit: 1.0**

Field work would be conducted any environmentally important area not less than 7 days. Each student shall submit a written report and present a seminar group-wise on the work done.

Following issues would be considered during the field work and in preparation of field report:

- Natural resources (water, land, minerals, flora and fauna, solar and wind energy) and their utilization pattern and future prospects.
- Natural and man-made pollution including industrial and agrochemical pollution.
- Eco-system sustainability.

- Public health and sanitation.
- Natural hazards and disaster.
- Environmental awareness and rights status and role of government and non-government organization.
- Environmental Impact Assessment.
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ESE 3206. Viva Voce**Credit: 1.0**

Comprehensive oral examination of the courses taught.

ESE 4101. Flood Control and Watershed Management**Credits: 2; Contact Hours: 2 Hours/Week**

Floods: causes, types, control methods.

River Engineering: River characteristics; Sediment discharge; Classification of Rivers and history; Sediment transport mechanism; River flow characteristics; Bed theory; River control; Estuary mechanics; Saline water intrusion; National and international issues on River water; Dredging of River bed; Rivers, the behavior, Control and training; Port theory; Dam and bridges on River; River pollution; Irrigation purposes of river; Runoff, infiltration; river guide theory. Geo-morphology: Channel: development, widening, Valley shape; Stream terraces; alluvial flood plains; Delta and alluvial fans; Channel morphology; Channel pattern and river basin; Generalized and load characteristics of river channel; Bed forms in the alluvial channel of sediment transportation on the river morphology and channel pattern.

Concepts and important features of a watershed, watershed management as a component of environment; nutrient and hydrological cycle in forest ecosystem, precipitation, interception, infiltration, percolation, ground water, surface runoff, stream flow and their measurements in forested and other land use systems in hilly watershed; role of forest on precipitation, falling rain (interception) infiltration and water storage on upland watershed, stream density, drainage density, average slope of watershed and mean elevation of small and large watershed, erosion rates under different land uses and sediment load estimation methods with relative advantages and disadvantages, control measures of gully and shifting cultivation in hilly areas, planting and forest management techniques in steep sloped watershed.

Recommended References:

- FAO/UNDP, 1971. Bangladesh Soil Resources, Soil Survey Project, AGL: SF/ PAK 6 Technical report 3, p. 185-
- Hewlett, K.D. 1982. Principles of forest Hydrology. The University of Gorgia press Athens, 184 pp. Hundson, N. 1971. Soil Conservation. B.T. Bastford Limited. 320 pp.
- Kohnke, H and A. R. Bertrand. 1959. Soil conservation. McGraw Hill
- Book Co. New York. 298pp. Negi, S.S. 1983. Soil Conservation Fundamentals of Forestry. Vol. 401. Bishen Singh Mahendral Singh. 23-4, Connaught place, Dehra Dun. 88p.
- Raeder-roitzsch, J.E. 1968. Lectures on Watershed Management and Forest Hydrology, Pakistan, Forest College, Peshwar. 244pp

ESE 4103. Environmental Chemistry**Credits: 2; Contact Hours: 2 Hours/Week**

Introduction to Environmental Chemistry: Introduction and concept of Environmental Chemistry, Components of Environmental Chemistry, Significance and role of Environmental Chemistry for Environmental science and resource management

Chemistry of lithosphere: The composition of the crust, the geochemical classification of the elements, distribution of elements in igneous rocks, distribution of elements in different types of rocks.

3. Chemistry of the atmosphere - the composition of the atmosphere, the evolution of the atmosphere, constancy of atmospheric composition:

Chemistry of hydrosphere: The composition of sea water, the composition of terrestrial water, gain and loss in the ocean, the balance of dissolved matter in seawater, the history and evolution of the ocean, the composition of the atmospheric water; the environmental significance of the chemistry of hydrosphere.

The chemistry of biosphere: The mass of the biosphere, the composition of the biosphere, the biogenic deposits, the origin of coal and petroleum: Some significant chemical processes - photosynthesis, fermentation, biodegradation etc

Chemistry of pollutants: Surfactants, pesticides, sewage etc:

Recommended References:

- Holland HD (1978) The Chemistry of the Atmosphere and Oceans.
- Wiley Interscience, NY Berner EK and Berner RA (1987) The Global Water Cycle – geochemistry and environment.
- Prentice Hall O’Niell P (1998) Environmental Chemistry.
- Blackie Drever JI (1988) The Geochemistry of Natural Water. Prentice Hall, NJ Mason B and Moore CB (1982)
- Principles of Geochemistry. John Wiley and Sons Krauskopf KB and Bird DK (1995)
- Introduction to Geochemistry. McGraw-Hill International

ESE 4102. Environmental Chemistry- Lab**Credits: 1.5; Contact Hours: 2 Hours/Week**

Preparation and standardization of acid and base solution.

Determination of major physical and chemical properties of water and wastewater (pH, EC, turbidity, TDS, TSS, total solids, total hardness, total alkalinity, total acidity, DO, BOD, COD).

Determination of the concentration of major metal ions in water and wastewater (Fe, Cu, Pb, Cr, Ni, Ca, Al).

Complexometric titration with EDTA as primary standard substance (using Murexide and Solochrome Black/Eriochrome Black T indicator).

Gravimetric analysis of some major anions (chloride, carbonate, sulphate, phosphate).

Introduction: Definition of environmental engineering

Environmental System Overview: Water resource Management system, Air resource management system, Solid waste management system

Water Supply, Sanitation and Wastewater Engineering: History and Development of water supply system, Elements of water supply system, water quantity and quality, Characteristics of water: physical, chemical and biological parameters, standard methods of water analyses; Introduction to sanitation, technological options, storm water and sullage drainage; Water, Human waste and wastewater treatment plants and systems: (physical, chemical and biological systems, primary, secondary and tertiary treatment, sedimentation, coagulation, flocculation, filtration, adsorption, ammonia removal, aeration, anaerobic and aerobic digestion, activated sludge and trickling filter, ion exchange, lagoons, disinfection, natural treatment systems); Water distribution systems, storage reservoirs and wastewater collection systems, pipe networks and network design, sanitary sewer and storm water collection system design, basic pumps/prime movers, urban drainage and runoff control.

Air Quality Monitoring and Pollution Control Engineering: Air quality: Definitions, characteristics and perspectives; Dispersion and deposition modeling of atmospheric pollutants (Eddy and Gaussian diffusion models, Puff models, effective stack heights and spatial concentration distributions, Measurement techniques; Control of pollutant particulates: collection mechanisms and efficiencies; Control of gases and vapors: adsorption, absorption, combustion, incineration; Control of sulphur oxides and oxides of nitrogen; Air toxics, mobile sources of air pollutants, noxious pollutants, and odor control.

Solid Waste Management Technology: Engineering design and operational aspects of waste generation, collection, storage, transfer, processing, including composting of organic waste, treatment and disposal; Engineering evaluation of: integrated waste management, solid waste characterization and classification, reduction, reuse and recycling, resource recovery and utilization; Life cycle assessment of waste, physical and chemical treatment methods and composting; Landfill design and operation including: site selection, engineered sites, liners and covers, leachate control and treatment, gas recovery and control.

Recommended References:

- Ahmed, M. F. and M. M Rahman. Water Supply and Sanitation. ITN-Bangladesh, Centre for Water Supply and Waste Management, BUET, Dhaka, Bangladesh.
- American Water Works Association. 1990. Water Quality and Treatment- A Handbook of Community
- Eddy, M. 1996. Wastewater Engineering. McGraw-Hill Inc.
- Peavy, Rewe Technology lous (1985); Environmental Engineering
- C.S. Rao; Environmental Pollution Control Engineering
- V. P. Kudesia; Industrial Pollution
- Gurnham, C. F. Principles of Industrial Waste Treatment. John Wiley and Sons Inc. New York
- S. P. Mahajan; Pollution Control in Process Industries
- Hammar, M. J. and M. J. Jr. Hammar. 1996. Water and waste water technology. Prentice Hall Inc
- Huda, K. M. N. Municipal Solid Waste Management- Bangladesh Perspective.
- Pontians, F. W. Water Supplies. McGraw-Hill Inc.
- Rowe, P. and Tchobanoglous. Environmental Engineering, McGraw-Hill Inc.

ESE 4107. Global warming, Climate change and Adaptation

Credits: 3; Contact Hours: 3 Hours/Week

EARTH'S CLIMATE SYSTEM -Introduction-Climate in the spotlight - Climate Classification - Global Wind Systems –Cloud Formation and Monsoon Rains – Storms and Hurricanes - The Hydrological Cycle – Global Ocean Circulation – El Nino and its Effect - Solar Radiation –The Earth's Natural Green House Effect – Green House Gases and Global Warming – Carbon Cycle.

OBSERVED CHANGES AND ITS CAUSES -Observation of Climate Change – Changes in patterns of temperature, precipitation and sea level rise – Observed effects of Climate Changes – Patterns of Large Scale Variability – Drivers of Climate Change – Climate Sensitivity and Feedbacks – The Montreal Protocol – UNFCCC – IPCC –Evidences of Changes in Climate and Environment

IMPACTS OF CLIMATE CHANGE- Impacts of Climate Change on various sectors – Agriculture, Forestry and Ecosystem – Water Resources – Human Health – Industry, Settlement and Society – Methods and Scenarios – Projected Impacts for Different Regions– Uncertainties in the Projected Impacts of Climate Change – Risk of Irreversible Changes.

CLIMATE CHANGE ADAPTATION AND MITIGATION MEASURES- Adaptation Strategy/Options in various sectors – Water – Agriculture – Infrastructure and Settlement including coastal zones – Human Health – Tourism – Transport – Energy – Key Mitigation Technologies and Practices – Energy Supply – Transport – Buildings – Industry – Agriculture – Forestry - Carbon sequestration – Carbon capture and storage (CCS)- Waste (MSW & Bio waste, Biomedical, Industrial waste – International and Regional cooperation.

CLEAN TECHNOLOGY AND ENERGY-Clean Development Mechanism –Carbon Trading- examples of future Clean Technology – Biodiesel – Natural Compost – Eco- Friendly Plastic – Alternate Energy – Hydrogen – Biofuels–

Recommended References:

- Asian Development Bank. 1994. Climate Change in Asia: Bangladesh Country Report, Published by ADB.
- Hug, S. 1999. Vulnerability and Adaptation to Climate Change for Bangladesh, Kluwer Academic Publishers.
- Islam, T. and A. Neelim. 2010. Climate Change in Bangladesh. The University Press Limited, Dhaka, Bangladesh.
- Washington, W. M. and C. L. Parkinson. 1986. An Introduction to Three Dimensional Climate Modeling, USA.
- Lockwood, J. G. World Climatology: An Environmental Approach. Edulard Arnold.
- Jan C. van Dam, Impacts of “Climate Change and Climate Variability on Hydrological Regimes”, Cambridge University Press, 2003
- Al core ‘inconvenient truth’ – video form
- IPCC Fourth Assessment Report – The AR4 Synthesis Report,
- Dash Sushil Kumar, “Climate Change – An Indian Perspective”, Cambridge University Press India Pvt. Ltd, 2007

ESE 4109. Natural Resource Management

Credits: 2; Contact Hours: 2 Hours/Week

Introduction: Definition, Concept, Importance of resources, Classification of resources, Resource use and Management, definition and emergency of sustainable development, sustainability and resources. **Overview of Water Resource and their Management:** Water as a resource, Status of water resources-Goal and regional perspective, Sustainable principles for water resource management, stakeholder’s involvement in water resource management, smart approach in water saving, Integrated Water Resource Management (IWRM): Concept, Components, Principles and importance, Application of appropriate technology for conservation of water resources of Bangladesh **Forest Resource and their Management:** Forest as resource, thematic elements of sustainable forest management (Extent of forest resources, Forest biological diversity, Forest health and vitality, Productive functions of forest resources, Protective functions of forest resources, Socio-economic functions of forests and Legal, policy and institutional framework), Bangladesh perspective. **Energy Resource Management:** Energy as resource, energy use, renewable and non-renewable energy resources, energy efficiency, energy conservation and its importance, energy saving techniques, energy and sustainable development, Bangladesh perspective. **Participatory Natural Resource Management;** Participatory approaches, Natural Resource Management (NRM) extension, participatory assessment and planning, participatory appraisal methods, monitoring and evaluation, capacity building. **Environmental and Resource Use Challenges:** Water scarcity, abuse of land resources, Energy crisis, Environmental effects of resource extraction,

Recommended References:

- Cunningham and Cunningham (2006). Principles of Environmental Science: Inquiry and Applications (3rd edition). McGraw Hill Publishers.
- Miller, G.T. (2004). Environmental Science-working with the Earth (10th edition). Jack Carey publisher.
- Joseph Priest, Energy Principles, Problems and Alternatives, 3rd Edition, Addison-Wesley Publishing Company.
- Devid Reed, Editor, Structural Adjustment, the Environment, and Sustainable Development, Earthscan Eastern Publications Ltd. London.
- Falkenmark, M., Andersson, L., Castensson, R., Sundblad, K. in collaboration with Batchelor, C., Gardiner, J., Lyle, C. Peters, N., Pettersen, B., Quinn, P., Rockström, J., and Yapijakis, C. (1997). Water a reflection of land use- Options for counteracting land and water mismanagement. Swedish Natural Science Research Council.
- Lundqvist, J., C. de Fraiture and D. Molden (2008). Saving Water: From Field to Fork – Curbing Losses and Wastage in the Food Chain. SIWI Policy Brief. Stockholm International Water Institute (SIWI).
- Nyambod, E.M. and Nazmul, H. (2010). Integrated Water Resources Management and Poverty Eradication –Policy Analysis of Bangladesh and Cameroon. *J. Water Resource and Protection*, pp. 191-198
- Global Forest Resource Assessment (2010). Key Findings. Food and Agriculture Organization (FAO) of the United Nations.
- Herring, H. (2006). Energy efficiency-a critical view. *Energy* 31, pp. 10-20.

Introduction: Scope of Environmental Modelling, the need for environmental modelling; Modelling elements and model classification– physical model, mathematical model; Software and program based studies on environmental problems; Computer modelling; Mass Balances, Model Calibration and Verification.

Transport Phenomena: Introduction, Advection, Diffusion and Dispersion, Compartmentalization, Sediment Transport, Simple Transport Model.

Model for Groundwater Contamination: Introduction, Darcy's Law, Flow Equations, Contaminant Solute Transport Equation, Sorption, Retardation and Reactions, Mobilization Models, Biotransformation, Bio-film and Bio availability, Unsaturated Zones, Remediation, Numerical Methods.

Three-dimensional climate modeling: Physical description of the climate system, Basic model equations, Basic methods of solving Model equations, Examples of simulations of present Day climate; Vulnerability, Adaptation and Societal Response to Climate Change; Policy and Framework Convention on Climate Change.

Global Change and Climate models: Climate Change and Global Circulation Model (GCMs; Global Climate Models or General circulation models), Global Carbon Box Model, Zero-dimensional models, Radiative-Convective Models, Higher Dimension Models, EMICs (Earth-system Models of Intermediate Complexity).

Model simulation: Process, schematic diagram, validation and robustness.

Recommended References:

- Environmental Modelling, Jerald L. Schnoor, John Wiley & Sons, 1996.
- Transport Modelling for Environmental Engineers and Scientists, Mark M. Clark, John Wiley & Sons, 1996.
- Energy and Environment - Modelling and Simulation, B.K. Bala, Nova Sci Pub, USA, 1998.
- Principles of System Dynamics, B.K. Bala, Agrotech Pub Academy, India, 1998.
- Introduction to Ecological Modelling – putting theory into practice, M. Gillman, R. Hails, Blackwell Science, 1997.

Models: Population, Physical, Mathematical and Management model, mechanism of pollutant fate in the environment

Air Quality Modeling: Mathematical models for air quality control and their application.

Ground Water Modeling: Introduction to existing ground water models.

Surface Water Modeling: Hydrodynamic modeling of rivers, water quality modeling of riverine system, modeling of water quality in estuaries, lakes and reservoirs.

Urban Water Supply Modeling: Introduction to existing urban water supply models.

Watershed modeling: Introduction to existing watershed models

Modeling of Wastewater Treatment: Introduction to existing models of wastewater treatment.

Modeling of Environmental Media: Introduction to employ the existing model related to environmental media such as Soil, Water and Atmosphere.

Environmental Analysis and Laboratory Management- Quality control and quality assurance.

Spectroscopy: The nature of radiant energy, interaction of light with matter, laws of photochemistry, quantum yields, wave mechanics, atomic and molecular absorption, origin of spectra, absorption and emission spectra.

UV-Visible Spectroscopy: Basic principles of spectroscopy, color detection, wavelength variation, absorption spectra of functional groups, instrumentation, absorption standards, stray correction, spectra-structure correlation, application.

Atomic Absorption Spectroscopy: Theory of AAS, atomization techniques, methods of calibration, standard addition, scope, sensitivity and application of ICP-AAS, ICP-AES, ICP-MS techniques in environmental analysis.

Chromatographic Method: Principles of chromatography, instrumentation and applications of chromatography, chromatographic optimization, matrix preparation, HPLC in biological and forensic analysis, chromatographic separation of DDT and PCBs.

Other techniques: Thermogravimetric methods (TG), X-ray Emission, X-ray fluorescence and electroanalytical techniques; potentiometry, coulometry.

Research methodology: Introduction- need and importance of research methodology

Methods of data collection – primary data, secondary data, selection of appropriate data, guidelines for interviewing, questionnaire and schedule, sampling protocols for water, soil, air and biological species, preservation of samples
Methodology and limitation of research.

Measurement and Scaling Techniques - problems, types, preparing of data for analysis, data interpretation and speculation. statistics, interpretation of results, graphical representation and tabulation.

Manuscript Preparation: Rules of quotation and bibliography

Report Writing: Scientific reports, Reviews, short communication, Case studies, web page, booklet and leaflet.

Recommended References:

- Islam, M. N. 2008. An Introduction to Sampling Methods. Mullick and Brothers. New Market, Dhaka.
- Islam, M. N. 2009. An Introduction to Research Methods. Mullick and Brothers. New Market, Dhaka.
- Kumar, R. 2005. Research Methodology. Pearson Education, India.
- Spectroscopic Method in Organic Chemistry, D.H. Williams, I. Fleming, 4th Edn, Tata McGraw-Hill, 1988.
- UV Spectroscopy, Techniques, Instrumentation, Data Handling, B.J. Clark, I. Frost, M.A. Russell, Chapman & Hall, 1993.
- High Performance Liquid Chromatography, Fundamental, Principles and Practice, W.J. Lough, I.W. Wainer (Edn), Blackie Academic & Professional, 1991.
- Understanding Instrumental Analysis, J.W. Robinson, 5th Edn, Marcel Decker, Inc., 1995.
- Atomic Absorption Spectroscopy, Theory, Design & Application, S.D. Haswell (Edn), Elsevier, Vol 5, 1991.
- Infrared Spectral Interpretation: A Systemic Approach, B. Smith, CRC Press, 1999.
- Essential Environmental Science: Methods and Techniques by Simon Watts

ESE 4201. Water Supply and Sanitation

Credits: 3; Contact Hours: 3 Hours/Week

Water supply

Sources of Water: Sources of Water in Bangladesh, Some important water quality parameters, Drinking water quality standards.

Water Supply Basics: Importance of water supply, History and development of water supply, Objectives of water supply, Elements of water supply, Planning and design considerations; population prediction and water requirement, collection system and intake design, water pipes and pipe fittings including valves, washout, hydrants etc.; cost and billing of piped water supply, water supply system in rural and urban areas.

Water Transmission and Distribution: Classification of transmission and distribution System, Distribution network, Design considerations, Service connections.

Alternative Water Supply Technologies: Shallow Shrouded Tube-well (SST), Very Shallow Shrouded Tube-well (VSST), Pond Sand Filter (PSF), Rain Water Harvesting, Infiltration Gallery, Solar Desalination.

Water Treatment Processes: Plain Sedimentation, Coagulation and Flocculation, Filtration, disinfection, Heavy Metal and Trace Elements removal (Arsenic, Iron, Manganese, Boron etc), Removal of Hardness.

Sanitation

Instruction: Dwellers sanitation; Code of practice for adequate environmental facilities in an infrastructure,

Housing and Buildings: Space, lighting, air movement and circulation, temperature control, plumbing facilities, ventilation and air conditioning, Principles of excreta disposal, Duckweed treatment for waste; Bio-gas plant; Different sanitation options; Sanitation practices in Bangladesh

Environmental Sanitation in Rural and Urban Areas: Environmental Sanitation, Rural and Urban Sanitation, Relationship between water, waste and health, Modes of transmission of diseases, Transmission routes of water and waste water related diseases, Control of diseases, Important measures to prevent transmission of diseases, Participatory management, Sanitation practices and problems in Bangladesh.

Recommended References:

- Ahmed and Jahan. Participatory Management of Low Cost Water Supply and Sanitation. Ahmed and Rahman. Water Supply and Sanitation
- Kapoor, B. S. Environmental Sanitation
- Punmia B. C., J. Ashok and J. Arun. Water Supply Engineering.
- Rangwala. Water Supply and Sanitary Engineering

ESE 4203. Environmental Planning and Design

Credits: 3; Contact Hours: 3 Hours/Week

Introduction: Significance of environmental planning and design in sustainable environment.

Concept of Planning: Development and developing countries.

Planning Process and Methodology: Content and function, the plan as a process, social and historical considerations, elements of planning and team work.

Design Plan and Environmental Design Fundamentals: Implementation process, comprehensive plan, Zoning plan, industrial performance, historical preservation, flexible zoning, specific plan; Geo-climatic factors in environmental design, Social and aesthetic.

Land use planning and Site Evaluation: Scope, importance, application of land use planning; Land use planning and management for commercial, residential agricultural, wetland areas etc and regulatory requirements; Impacts of natural and man-made activities on land use planning; Land use planning models and their limitations; Rules and Regulations for land use.

Energy Efficient Buildings: Concept, importance and applications, design considerations and examples from developed countries.

Introduction to urban planning: Historical thoughts of urban planning, Urban environmental matters, Urban management and brown agenda, Evaluative image of environment of a city/ town, Environmental disasters and hazards; GIS in urban environmental planning, Town and city planning, Transport planning, Environmental planning, Ecological planning. Urban environmental planning in Bangladesh: accelerating urbanization, selected urban planning project in Bangladesh.

Recommended References:

- Cowon, J. Environmental Design.
- Leboyer, C. L. Psychology and Environmental.

Principles and concepts of environmental biotechnology: Usefulness to mankind.

Degradation of high concentrated toxic pollutants: non-halogenated, halogenated -petroleum hydrocarbons- metals. Mechanisms of detoxification, oxidation reactions, dehalogenation- biotransformation of metals. Microbial cell/enzyme technology – adapted microorganisms - biological removal of nutrients – microalgal biotechnology and applications in agriculture- role of extra cellular polymers.

Biotechnological remedies for environmental damages: decontamination of ground water systems– subsurface environment- reclamation concepts- bioremediation. Production of proteins- biofertilizers.

Biodegradation of solid wastes: Physical, chemical and microbiological factors of composting- health risk- pathogens– odor management -technologies of commercial importance advances in biogas technology- case study.

Concept of DNA technology: plasmid- cloning of DNA- mutation- construction of microbial strains.

Environmental effects and ethics of microbial technology: safety of genetically engineered organisms.

Recommended References:

- Environmental Biotechnology, Concepts and Applications. Hans-Joachim Jordening and Josef Winter. Winter-VCH. 2005
- Biology of wastewater Treatment. N F Gray. Mc Graw Hill . 2004.
- An Introduction to Environmental Biotechnology by Milton Wain Wright. Kluwar Acad Publ. Group, Springer, 1999.
- Environmental Biotechnology – Sayler & Fox
- Principles of Gene Manipulation. 6th Edition, S.B.University Press, 2001- S.B. Primrose, R.M. Twyman and R.W.Old
- Fulker M.H. Environmental Biotechnology, CRC Press, 2010.
- Wainwright, M, An Introduction to Environmental Biotechnology, 1999.
- Martin, A.M., Biological Degradation of Wastes, Elsevier Appl. Science, New York,1991
- Gray, S.S., Fox, R and James W. Blackburn Environmental Biotechnology for Waste Treatment, Plenum Press, New York 1991.
- Rittmann, B.E, Seagren, E., Wrenn, B. A and Valocchi A.J, Ray, C and Raskin, L Insitu Bioremediation (2nd Ed.) Naves Publ. U.S.A. 1994.
- Old, R.W., and. Primrose, S.B., Principles of Gene Manipulation (3rd Ed.), Blackwell Sci. Pub, Cambridge, 1985.

ESE 4207. Industrial Ecology

Credits: 2, Contact Hours: 2 Hours/Week

Fundamentals of Industrial Ecology: Defining industrial ecology and its different perspectives. Industrial Ecology as a field of ecology, goals of industrial ecology. Key concept of industrial ecology system analysis, materials and flows and transformations. Multidisciplinary approach, analogies to natural systems, open vs. closed –loop system, the role of companies, dematerializations and eco-efficiency, industrial symbiosis.

History of industrial ecology: Agronomy and ecology, Ecological anthropology, different development of industrial ecology as a scientific discipline- The Belgium ecosystem, The Japanese view and American view, a brief history of cleaner production.

Material and energy consumption in context of industrial ecology: Energy resources and classification (renewable and non-renewable), production data, materials resources classification (renewable and non-renewable), global material resources, water and energy, consequences of living in an industrial world.

Life Cycle Assessment (LCA): Definition and applications of LCA, Historical development, LCA and ISO14000 standard, technical framework of LCA. Components and methodology of LCA, impact categories for life cycle impact assessment, the choice of model parameter and uncertainty in LCA, software and database for conducting LCA, specific uses of LCA, LCA of product and process for manufacturing and service sectors.

Emerging concept of industrial ecology: Sustainable resources and materials management in developed and developing countries, industrial metabolism and cannibalism, industrial scavenging, waste mining.

Industrial Ecology-Bangladesh perspectives: Applicability of industrial ecology in Bangladesh, cases of industrial ecology investigations in Bangladesh, state of industrial ecology future directives.

Recommended References:

- Robert U. Ayres and Leslie W. Ayres (edt.), A Handbook of Industrial Ecology, Eward Elgar Publishing Limited, UK
- Ehrenfeld, John and Nicholas Gertler, Industrial Ecology in Practice: The Evolution of Interdependence at Kalundborg. Journal of industrial ecology (1997) 1(1): 67-79
- Environmental Life Cycle Assessment of Goods and Services: An Input- Output Approach, Chris T. Hendrickson, Lester B. Lave and H. Scott Matthews, Routledge.
- Environmental Life -Cycle Assessment, Marry Ann Curran, McGraw-Hill
- Handbook of Input-Output Economics of Industrial Ecology, Sangwon Suh (Edt.), Springer
- Industrial Ecology and Global Change, R. Socolow, C. Andrews, F. Berkhout, and V. Thomas (Edt.), Cambridge University Press, UK
- Industrial Ecology, T.E. Graedel, and Braden R. Allenby, Prentice Hall, ISO14000 International Standard, Environmental Management-Life cycle assessment-Principles and framework, 1997, 06-15
- Jelinski, L.W, T.E. Graedel, R.A. Laudise, D.W. McCall and C. Kumar N. Patel. Industrial Ecology: Concept and approach. Proceedings, National Academy of sciences, USA 89 (1992): pp.793-797
- Life Cycle Assessment in the Built Environment, Robert H Crawford Taylor & Francis.

Introduction to public health engineering, the scope of public health engineering, principles of public health engineering research, public and environmental health–current issues, Epidemiology public health measures for disease control, Disease emergence and reemergence, toxicology, Vector-borne disease, Vector-borne disease control methods, common source diseases – food and water, health aspects of excreta disposal, on-site wastewater treatment, water demand and supply, emerging diseases, common source disease, drinking water supply and reuse, excreta and waste management, integrating medicine and public health, rural and urban water supply engineering, sanitary engineering, planning and budgeting in public health, nanotechnology health and safety, public health infrastructure – local, regional and national levels, water supply and sanitation sector status in Bangladesh, Global public health

Recommended References:

- Earle B. Phelps, Public Health Engineering: A Textbook of the Principles of Environmental Sanitation, Volume 1
- K.N. Duggal, Elements of Public Health Engineering, Publisher: Chand, 1967
- Gute, D.M. and Hanes, N.B. An applied approach to epidemiology and toxicology. NIOSH: Washington, D.C., 1993. Unit IV.
- Pencheon, David; Guest, Charles; Melzer, David; Gray, JA Muir (2006). Pencheon, David, ed. Oxford Handbook of Public Health Practice. Oxford University Press
- Smith, Sarah; Sinclair, Don; Raine, Rosalind; Reeves, Barnaby (2005). Health Care Evaluation. Understanding Public Health. Open University Press.
- Porter, Dorothy (1999). Health, Civilization, and the State: A History of Public Health from Ancient to Modern Times.
- P.N. Modi (2006), Water supply Engineering – Environmental Engineering (Vol.I)
- S.K. Garg (1999), Water supply Engineering – Environmental Engineering (Vol.I)
- Rosen, George. A History of Public Health. New York: MD Publications 1958
- Turnock, Bernard (2009). Public Health: What It Is and How It Works (4th ed.). Sudbury, MA: Jones and Bartlett Publishers.
- White, Franklin; Stallones, Lorann; Last, John M. (2013). Global Public Health: Ecological Foundations. Oxford University Press

ESE 4211. Water Pollution and Quality Management**Credits: 2, Contact Hours: 2 Hours/Week**

Water Pollution: Definition, major sources of water pollution, Nature and types of water pollutants: Heavy metals, organic-inorganic substances, pesticides, PCB and Radionuclides, Effects of water pollution on biota.

Pollution of Surface water: Sources of Surface Water Pollution; Metals as Surface Water Contaminants; Organic Compounds in Water; Vulnerability and states of pollution of surface water.

Groundwater Contamination: Point and diffuse Source Contamination; Enteric Pathogens as Surface Water Contaminants; Sources of contamination, Types of Contaminants; Sea-water Intrusion, Radioactive heavy metal and trace elements contamination.

Water Quality Criteria: Quality Assessment, Monitoring and Protection

Remediation of Contaminated Groundwater: Soil Remediation, Groundwater Remediation.

Water quality standards: Bangladesh water quality standard, International water quality standard-WHO, EU, etc, measures of water quality-different water characteristics; natural water quality; polluted waters.

Emergence of water quality issue: important characteristics of water; availability of water as resources; practical environmental implications in water uses

Causes of Water Quality Problems: Sources and quality of water pollution type, drivers of potential pollutants and polluting agents causing water quality deterioration; evaluation of stream pollution

Major Water Quality Issues in Bangladesh: Evaluating risk of water quality vs. human health, Arsenic contamination and management; Water contamination by flood and its management; Industrial effluents in water quality deterioration; Agrochemicals from water quality perspective.

Indigenous techniques in water quality management in Bangladesh: Rain water harvesting, Boiling and settling in pitchers, Application of CaO and alum; Sustainable use of ground and surface water, law and policies related to water quality.

Recommended References:

- Lamb, J.C. 1985. Water Quality and its Control.
- John Wiley & Sons, N.Y., Rahman, A. A. *et al.* (eds.). 2000. Environmental Aspects of Surface Water Systems of Bangladesh. The University press limited. 114 Motijheel C/A. Dhaka.
- Steel, E.W. and Mc Ghee, T.J. 1985. Water Supply and Sewerage McGraw Hill Book Company, New Delhi. Trivedi, P.R. and Raj, G. 1997. Water Pollution. Akashdeep Publishing House, New Delhi-110002,
- Linsley, R.K., Kohler, M.A. and Paulhus, J.L.H. Applied hydrology.
- U.S. Department of Interior Ground Water Manual
- Walton, W.C. Ground water resources evaluation
- UNDP (1982) The Hydrogeological Conditions of Bangladesh.
- Mathess, G. (1982) The Properties of Groundwater.

ESE 4202. Water Pollution and Quality Management –Lab**Credits: 1.5, Contact Hours: 3 Hours/Week**

Determination of Physical, Chemical and Biological properties of surface water/groundwater (BOD, COD, DO, pH, Electrical conductivity, TDS, Salinity, Turbidity, Hardness, Dissolved solid, Suspended Solid, Heavy metals, Trace elements, etc.), Interpretation of Water Chemistry Data.

ESE 4204. Research / Project Works**Credits: 3+1, (Report + Presentation)**

- A research/ project works on environmental issues under the supervision of any faculty of the department and/or an internship in any reputed organization can be undertaken for 3 months as assigned by the Department. A report must be submitted by each student and the report is to be evaluated by internal and external member as assigned by the Department.
- Students may allow to choose their research/project work with corresponding supervisor. Review of relevant literature or scientific publications on a selected issue in Environment and /or conduct of simple research/ study on a Environment problem. The text of the review paper topic should not more than 100 pages (spacing 1.5, font size 12) including references.
- Students are required to make up a time-frame plan for the work
- Students are required to discuss the work plan with his/her supervisor
- Students are required to submit a report with literature study/search for facts; Planning the experimental work/method description and Experimental work/ theoretical studies and analysis of results and data
- Students are required to submit his/her report to the supervisor; after improvement draft report can be submitted to the department
- Students are required to appear before his/her report in the seminar/presentation prescribed by the department
- Students are required to submit minimum 4 copies of his/her final report to the department (i. Seminar library copy ii. Central library copy iii. Supervisor copy & copy iv. Student copy), in case of co-supervisor, more copy may has to be submitted.

Review and Evaluation of thesis/report:

Logic and organization: Problem well defined, rationale clearly stated; Methods clearly described; Data collection, analysis and guided an appropriate conceptual framework; Conclusions/Recommendations based on observation and/or analysis. (40%)

Relevance: Topic is directly related to current Environment issues; Materials/ Information presented is appropriate. (15%)

Simplicity and clarity: Ideas are expressed clearly and simple; correct grammar used and rules of composition observed. (15%)

Originality and creativity: Diverse information/ideas are well integrated or synthesized into a new form/ structure; Interesting and relevant insights are drawn from study results. (30%)